A STUDY ON FISH STOCK DEPLETION WHICH THREATEN LIVELIHOODS ON FISHERIES SECTOR IN KERALA WITH SPECIAL REFERENCE TO VEMBANADU LAKE, KUMARAKOM

Dissertation submitted to Mahatma Gandhi University, Kottayam

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DECLARATION

I hereby declare that the dissertation entitle "**A STUDY ON FISH STOCK DEPLETION WHICH THREATEN LIVELIVEHOODS ON FISHERIES SECTOR IN KERALA WITH SPECIAL REFERENCE TO VEMBANADU LAKE, KUMARAKOM**" is exclusively a bona fide record of research work done by me under the supervision of **Asst. Prof. Dr. Sr. DEEPA K THOMAS, Department of Economics, Bishop Kurialacherry College, Amalagiri** and is submitted to the Mahatma Gandhi University in partial fulfillment of the requirement for the Degree of Masters of Arts in Economics and that it not previously formed the basis for the award of any Degree, Diploma, Fellowships, or any similar title of any other University or institution.

Place: Amalagiri

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CERTIFICATE

This is to certify that, the dissertation titled " **A STUDY ON FISH STOCK DEPLETION WHICH THREATEN LIVELIHOODS ON FISHERIES SECTOR IN KERALA WITH SPECIAL REFERENCE TO VEMBANADU LAKE, KUMARAKAM**" , is a bona fide work done by **Ms. DEVI RAMESH**, Department of Economics, Bishop Kurialacherry College, Amalagiri, in partial fulfillment of the requirements for the award of the Master of Arts in Economics in Mahatma Gandhi University under my supervision and guidance. This work was not previously formed the basis for the award of any Degree, Diploma, Fellowship, or any similar title of any other university or institution. She is permitted to submit the dissertation.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Fishery can mean either the enterprise of raising or harvesting fish and other aquatic life; or one commonly, the site where such enterprise take place commercial fisheries include wild fisheries and fish farms, both in freshwater water bodies and the oceans. About 500 million people worldwide are economically dependent on fisheries. 171 million tonnes of fish were produced in 2016, but overfishing is an increasing problem causing declines in some populations.

Because of their economic and social importance, fisheries are governed by complex fisheries management practise and legal regimes that very widely a cross countries. Historically, fisheries were treated with a "fish-served" approach, but recent threats by human overfishing and environmental issues have required increased regulations of fisheries to prevent conflict and increase profitable economic activity on the fishery. Modern jurisdiction over fisheries is often established by a mix of international treaties and local laws.

Kerala is blessed with an estimated marine fishery potential of a remarkable 5.17 lakh tonnes, the department has can surprisingly a major role to play in the economy of the state. In addition, the department carries out a number of programmes and project the welfare of the fisher folk. Fisheries sector plays crucial role in the wellbeing of Kerala's economy. The Department of fisheries is headed by the Director of fisheries. The Department of fisheries in the state is structurally stratified and organized under the following executive officers. Among the maritime status in India Kerala occupies the foremost position in marine fish production, according for about 20% of the total landings.

Vembanad Lake is a transitional ecotone between sea and land having a length of 96 km and a surface area of 1512 km^2. The rich biodiversity and socio economic importance of Vembanad Lake along with adjacent kole lands led to

the declaration of the lake as a Ramsar site, a wetland of international importance. Ten rivers which originate from the Western Ghats biodiversity hotspot drain to the lake and eventually join the lake like pathiramanal also provide habitat for resident and seasonal migratory water fowls, others, fish, clams, shrimps, crabs, aquatic insects and other aquatic organisms. The lake is also famous for its tourism potential, live clam resources and sub-fossil shell deposits, as a habitat for the vulnerable spot billed pelican pelicances philippensis, large population of water fowls, besides a high species of fish belonging to 100 genera and 56 failies are known to occur in Vembanad Lake. Extensive water and remodelling efforts in Vembanad Lake has drastically changed the lake's water and landscapes, altering natural habitats of many terrestrial and aquatic fauna and flora.

Vembanad Lake, India's largest is a key tourist attraction and source of livelihood for thousands of fisherman families in kumarakom and nearby panchayats . However, the dwindling fish wealth has plunged fisherman into crisis. There are 1400 fisherman registered with the fisheries department at kumarakom. Thiruvarppu and Arpukara panchayats in Kottayam districts of them, less than 50 percent fishers go fishing now. Most have left as the money is not good.

For those still in it, the inland fisherman welfare service cooperative society is a big relief as it procures fish at a good price. However, president of the society KK Rajappan says inland fishermen are facing several crises and require government's support. Fish production has come down drastically. As per a study, Vembanad Lake had around 172 species of fish before the thanneermukkom regulator was constructed in 1975. Recent studies say there are only 52 species now; he says. He blames the unscientific management of the regulator for this.

The barrage remains closed from December 15 to March 15 to prevent entry of saline water and help paddy cultivation. The regulator was constructed to aid in paddy cultivation, but it is happening just once a year mostly in kuttanad and upper kuttanad. With the regulator closed, breeding of fish has been affected its closure also affects the tide, which is vital for the fish population.

VEMBANAD FISH COUNT-REPORT OF THE PARTICIPATORY FISH RESOURCE SURVEYS OF VEMBANAD LAKE (KERALA) DONE IN 2008 AND 2009 BY ATREE

The vembanad estuarine system, the largest of its kind on the west coast of India is known to be abundantly enriched with diverse fishery resources, providing feeding, spawning and rearing areas for a very large proportion of commercial fish and shellfish. However, ecosystem health of the vembanad wetlands is alarmingly declining due to a variety of reasons – obstruction of river courses, sand mining & habitat destruction, loss of riparian canopy cover, encroachment, pollution and unethical fishing practices. Reduced summer flow due to drying up of rivers and pollution hazards from agro-chemicals and sewage also lead to mass mortality of fishes. Depletion of fishery resources has changed this ‘inland fish basket’ to an ‘inland wastebasket’, driving the fisherflock, the primary stakeholders of the lake into a livelihood crisis.

Traditionally, conservation and management of the lake resources of vembanad were vested in the hands of the local communities and hence, local communities were activity involved and interested in the management of the resources they relied on. Ashoka Trust for Research in Ecology and Environment (ATREE) is trying to work towards a deliberative, democratic conservation strategy for the conservation of the Vembanad wetland, so as to ensure the participation of the communities in conservation of the lake.

With the intention of the participatory methods in lake management and to enhance cooperation between various stakeholders, Government department , NGOs and academic institutions and to create awareness about the state of fishery resource of the lake, ATREE in collaboration with Regional Agricultural Research Station (RARS) Kumarakom, Kerala State Biodiversity Board, School of Environmental sciences – Mahatma Gandhi University Kottayam, School of Applied Life Sciences Pathanamthitta, St Albert's collage Ernakulam, Collage of Fisheries Panangad, Vembanad Nature Club Muhamma and various Kayal Samrakshana Samithyas (Lake Protection Forums) around vembanad, attempted to conduct a "Vembanad Fish Count" in 2008 and 2009, a participatory fishery resource inventory and monitoring of the Lake, involving scientists, naturalistic, students, fishermen, and interested people. Besides inventorisation, the effort was also aimed to create mass awareness among the general public on conservation of Vembanad Lake through more sensible management practices.

FISH COUNT-2008

The first edition of the fish count was on 30th may 2008. 51 species of finfish representing 26 families and 35 genera, as well as 10 species of shell fish belonging to 6 families and 7 genera were recorded during this fish count. Out of this, two species of fishes are listed as 'Critically Endangered', four Endangered' and five species are 'vulnerable' according to IUCN Red data book. The sector around Pathirammanal Island was observed to be the richest sector (in terms of fish) of the total area surveyed.

FISH COUNT-2009

The second edition of the fish count was held on 26th and 27th may 2009, and involved a wider group of organisers and participants. 65 species of finfish and 14 species of shell fish were recorded. Island was observed to be the richest sector surveyed with about 50 fish species recorded.

In total, the survey recorded 13 more native fish species and 3 shell fish species to the fish fauna of the lake over the previous year's fish count. It also recorded two exotic species in the lake, for the first time. Amongst the newly recorded species, 8 were estuarine species which proliferated the lake due to the timely opening of the Thannermukkam salt Water barriers and the remaining 5 were typical riverine species. Three species –Indian shortfineel, Indian mottled eel and Walking catfish were found to have disappeared from the lake. This development has been caused by stake net operations and dams in the northern part of the lake, which obstruct the migratory pathway of eels. The 2009 fish count report also points to a new threat to the native fish species of Vembanad Lake – Invasive Alien species such as the exotic sucker catfish.

1.2 SIGNIFICANCE OF THE STUDY

In India a number of people engaged in fishing and related employment is increasing those people have to suffer many problems like exploitation, poverty, illiteracy, ill health etc... More these at present they experienced a crucial situation.

The fishermen in kumarakom panchayath are in angry and facing difficulty due to fish stock depletion. The investigator is aimed on access the socio-economic condition of fishermen community in kumarakom panchayath through this, the family profile of fishermen, problem faced by the fishermen at the time of depletion of fish remedial measures by government and the ultimate source of live hood applicable to the area can be measured. And we can suggest remedial solution for their problem also.

1.3 STATEMENT OF THE PROBLEM

A lot of problems are faced by fishermen in Kerala. Illegal overfishing adds to concerns of fish stock depletion in Vembanadu Lake. As per estimate the

international research and training centre for below sea level farming, the overall fish wealth in the lake has come down from 23000 tonnes to about 4700 tonnes. Fisheries department has launched an exclusive patrolling across the region.

The rise of marauding fishing is just one among an array of anthropogenic activities that a depleting the resource of the vembanadu. The rest include pollution of the lake presents of the thanermukam barrage, an the loss of riparian vegetation. Therefore the preset study is giving to analysis fish stock depletion faced by fishermen and how to solve it.

1.4 OBJECTIVES OF THE STUDY

1. To examine the socio-economic conditions of the fisherman community in the study area.
2. To investigate the problems faced by fishermen due to fish stock depletion.
3. To identify the trend and structure of the fishery management practices in Kerala.
4. To suggest suitable policy measure to improve the conditions of fishermen society.

1.5 SAMPLE DESIGN

A sample size of 50 respondents has been selected from the 5 wards in kumarakom panchayat.

1.6 METHODOLOGY

This study is based on both primary and secondary data. The sample size is 50. Sampling procedure is random sampling and convenient sampling method. This tries to understand the fish stock depletion affects the fishermen in

Kumarakom panchayat, Kottayam. Well-structured questionnaires are used to collect primary data for this study. Secondary data collected from Matsya Bhavan, Kumarakom. The secondary data has also been collected from the Kerala Fisheries Department site and various articles, datas from kumarakom panchayath and internet etc.

1.7 LIMITATIONS OF THE STUDY

1. Limited area is one of the major limitations of this study.
2. Collection of primary data requires lot of time so this also limiting factor of the study.
3. Time and money are the major constraints of the study.
4. The study was restored with primary data and the unavailability of current recent data was a major constraints for this study.

1.8 TOOLS OF ANALYSIS

All the data were collected from the primary source has been classified, tabulated, or diagrammatically presented for more simplification, statistical tools like percentage, average etc have been used for analyzing the data.

1.9 CHAPTER SCHEME

This study is segmented into 5 chapters.

Chapter 1

Chapter 1 is an introductory chapter which deals with the Introduction, Statement of the problem, Significance of the study, Objectives, Methodology, Limitation, Tools of analysis, etc.

Chapter 2

Chapter 2 deals with Literature Review. It includes past studies referring to fish stock depletion.

Chapter 3

Chapter 3 deals with overview of the study.

Chapter 4

The chapter 4 deals with analysis of data and its results.

Chapter 5

The last chapter deals with findings and conclusions.

CHAPTER 2

REVIEW OF LITERATURE

REVIEW OF LITERATURE

1. Noble and Narayanan Kutty (1978) Indicated that the gross income in relation to investment is very good in the indigenous fishing units and gives proportionately higher rate of production than the mechanized units. The country crafts require comparatively less investment and it can be economically put into action even when the fish in the sea is scanty.
2. Sathiadhas and Venketaraman (1981) studied the impact of mechanised fishing on the socio-economic condition of the fishermen of sakthikulangar – Neendakara are of Kerala. This study found that there were improvements in housing, literacy, employment, infrastructure, production, exports and earnings of the region. It, however, recognized a rise in the level of indebtedness of the fishermen households which was attributed to the bank loans taken by the households for purchase of fishing vessels. The study further pointed out the lack of a fishing harbour as the major constrains affecting the development of the project area.
3. Sathiadas (1982) indicated that the gross and net earnings of motorized units increased due to higher catches of cuttle fish. However, he pointed out that there is not much difference in catch and revenue between motorized and non-motorized units along Tirunelveli coat where the wind blows favourable most part of the year enabling the non-motorized units to operate equally effective.
4. Bay of Bengal Programme (1982) reported that nearly 60 percent of the families go without any meal on some days. The major reason is low income due to low or no catch. As for food consumption vegetables, meat and milk are consumed occasionally, fish atlest half the year.
5. Kurien and Mathew (1982) in their study indicated that there are wide regional variations in the species-mix of the marine resource in the country. The nutritive

value of all species of fish is almost the same but there are different type of market demands and prices for various species. There is a direct impact of mechanization on fish production in quantitative as well as qualitative nature.

6. Balakrishnan and Alagaraja (1984) analysed and concluded that the reasons for the clashes between the mechanized boat owners and the indigenous craft owners caused to heavy damages in the year 1978 in the coastal areas from Jagathapattinam in Pudukkottai District to Mallipattinam in Thanjavur district. It was suggested that the forming of peace council comprising of the local R.D.O. as chairman and representatives from state Fisheries Department, machanized boat owners and indigenous craft owners as members worked satisfactorily to end the clashes. The peace Council drew schedules of restricted fishing for mechanized boats for three days in a week and traditional boats for four days. This was strictly kept up by the respective boat owners without any violation, while all the types of boats operate during day time. Since the system of regulated fishing worked satisfactorily, the same may be extended to other regions where the clashes between these two sectors exist.
7. Sam Bennet and Arumugam (1985) in their study concluded that quiet changes have taken place in the fishery due to the introduction of mechanization of traditional fishing crafts. The highlights are (a) Fishermen are benefited by increased catch per units as well as increased price for the catch by arriving earlier. (b) A significant improvement in the socio-economic structure of the traditional sector is the extension of mechanization to existing crafts and (c) number of persons in each boat has been reduced to three from five.
8. Bhavani (1986) in her report presents information on feeding and weaning practices and on health and nutritional problems among people in coastal villages surveyed. A considerable number of women and children suffered from partial blindness because of vitamin A deficiency. 70 percent of the children suffered from 1st and 2nd grade malnutrition. The most common diseases among children were respiratory infection, whooping cough and worms. Poverty and

irregular income, inadequate medical facilities, lack of knowledge and underutilization of facilities available were mainly responsible for these diseases.

9. Central Marine Fisheries Research Institute (1987) in its annual report stated that during January to March and up to mid of April fish landings which again slowly decline towards banned season. Therefore, the ban has thus a beneficial effect that helps in the conservation of the marine resources.
10. Institute for Techno-Economic Studies (1987) revealed that most of the families (62.6%) were with a debt average Rs.3100. As many as 58% had borrowed from moneylenders (at an interest rate of 36%), 13% from banks, and about 8% each from traders, co-operative societies and other relatives, respectively, while the rest had borrowed from boat owners. Clearly, the informal sector was the most important source of credit, while only about 20% of the credit was the accessed from formal sources. This is also linked to the fact that formal sources of credit are not adopted well enough to the realities of the sector, marking it more difficult to obtain and repay loans.
11. Sathiadas and Panikkar (1988) in their study analysed that the costs and earnings of traditional fishing units along Trivandrum coast, Kerala. The study covers catamarans with hooks and lines, catamaran with gillnets and plank-builts boat fitted with outboard motor (OBM). Considering the catch and revenue of different seasons for these unit, monsoon period (June-August) is found to be more productive and profitable. The study indicates that the catamaran unit shows better input-output and capital efficiencies as compared to outboard motor units since the initial investment is comparatively less. Catamarans with hooks and lines are highly suitable as a family enterprise for the small investors who can go fishing on their own units. However, in terms of higher productivity and gross and net income and employment potential, the boat fitted with OBM is more efficient.

12. Balan et al. (1989) conducted a detailed study on the impact of mechanization of country creaft in Kerala. The costs, earnings and key economic indicators for motorized and non-motorized plank-built boats, canoes and catamarans operating hook and lines, boat seines and gillnets were worked out. Returns to capital and labour were comparatively more for motorized units. Further, extending the area of operation and adopting diversified fishing methods become feasible due to motorization. It has been observed that motorization has brought an element of dignity to the fishing profession. The study indicated that the landings of motorized crafts have substantially increased ruing 1980 and non-motorized showed a declining trend.
13. Sathiadhas and panikkar (1989) studied that the cost-benefit relationship of three types of trawlers of sizes 8.5m, 9.5m. their initial investment, fixed cost, operating costs and returns were analysed. They had also calculated various key economic indicators for these types of trawlers. They concluded that the capital turnover ratio rate of return on capital and payback period were better for smaller boats (8.5m and 9m) in terms of labour productivity wages, quantum of catch, gross revenue and net profit, the bigger size boats (9.5m) were comparatively efficient.
14. Varambally (1990) studied that the performance of marine products in exports from 1981 to 1988 and had put forward some useful suggestions for further accelerating exports. According to him, the exporters have to adopt new marketing strategies to promote Indian marine products in other countries, understand the changing customers' preferences and scan the global market to identify strength, weakness, opportunity and threat to marine exports in international markets.
15. Sathiadas and panikkar (1991) in their study pointed out economics of catamaran fishing along the madras coast,, and concluded that the catamaran owners can enhance their earings by increasing the size of craft as well as number of gears. The poor economic condition coupled with limited availability

of finance from the institutional agencies force the fishermen to rely on less equipped fishing equipments, which in turn result in less returns entangling them in a vicious circle of poverty.

16. Radhasyam and Tripathy (1992) analyzed the experience of a village in Bhuvaneshwar of Orissa under composite fish aquaculture. In 1978, when the villagers were not aware of aquaculture. 88.8% of the families of this village lived below the poverty line with a monthly income of Rs.185 per family. After 1978, they adopted culture methods of rearing fish they started getting good returns. In 1990, when the survey was conducted, the monthly income of each family had increased to Rs.12863. The percentage of the family below the poverty line came down to 45.45%. The experience of this village shows that aqua culture diversified in many ways capable of contributing significantly towards a qualitative shifts in the socio economic status of the rural poor by enlarging the scope of self employment.
17. BOBP (1994) reported that fish was the main source of food and income for households in the rural coastal villages of Tamil Nadu. Fishing is the primary income generating activities in the rural coastal villages. A grater portion of the fisherfolk income goes for food. It was also reported that the fisherfolk among the rural coastal areas are mostly poor. Majority of them are engaged in fishing activity, but the income they earn is not enough for them to survive and to meet their expenditure. Due to this low level income, they are not able to send their children to school. Thus most of the fishermen are illiterate and only a minimum number have gone through primary education.
18. Mahesh V Joshy (1996) in his study had pointed out the importance of fisheries in Indian national economy. He analyses the impact of mechanization on fish catchment, income, export, standard of living and employment opportunities in marine the fishing sector
19. Bay of Bengal Programme (1997)conducted a nutrition survey in Nochikuppan and Ayodyakuppam marine fishing hamlets city which indicated that of 482

children under five years of age, 55% were malnourished mildly or moderately and 3% severely (Gomes weight/age classification). The families' dietary patterns showed quantitative and qualitative variations over the year, depending on the fishing seasons. 75% of the families spent Rs.10 to Rs.20 a day on food; while the daily income ranged from Rs.10 to Rs.30. Fish was the animal protein consumed almost every day by most families. Milk, meat and fruit were eaten rarely.

20. Kanwar Surendra Sing and Radhika (1996) in their study analysed socio-economic conditions of fisherfolk in Himachal Pradesh area. They laid stress on the comparative conditions of living of this section of the society. The applications of certain indices to the several aspects are well carried out in their micro work. But their findings could not throw light on the lacuna found so far as marketing channels and debt conditions are concerned.
21. Jacob Jerold Joel and Ebenezer (1996) in their study analysed that present status of trawls fishery at colachel for a period of five years from 1990 to 1994. The study indicated that the annual average gross income per fishing trip worked out to a minimum of Rs.8, 121 in 1990 to a maximum of Rs.22, 683 in 1994. Therefore, it was found that trawl fishing at Colachel was more productive and profitable.
22. Ali (1996) in his study suggested that the infrastructure facilities like health facility, clean drinking water, electricity, education are lacking among the rural coastal fishing communities. Most of the fishermen had no fishing assets like boats and nets for their own. Housing condition of the fisherfolk is not up to the expected level. Literacy rate and education are also low among them
23. Pathak (1997) in his study indicated that the availability of adequate and timely credit is the most crucial input for increasing the fish production and export. He also mentioned that the NABARD had so far sanctioned more than 5200 fisheries schemes with total refinance assistance of Rs.4037 million and the disbursement of credit for fisheries has increased at a compound annual growth

rate of 36.7 percent in the last ten years, when the aggregate assistance provided to all the scheduled banks reached a level of Rs.1071.7 million in 1995-96 from Rs.133.5 million in 1985-86. Thus the banker's contribution could best realised from the fact that out of the total marine fish landings of 2.469 million tonnes reported in 19992-93, more than 28 percent of the catch was from those boats that availed bank's credit facilities. In future, the bankers could take up fisheries programme and can invest over Rs.8000 million by the end of Ninth Five Year Plan and the seafood export would have a business of three billion US dollars in the international markets.

24. Government of Kerala (1997) concluded that higher cost and investment has also meant more indebtedness. Sources of credit are both the formal sector (banks and cooperatives) and the informal sector (trader, money lenders), through the latter continue to be more important. The findings of a survey by Matasyafed in Kerala revealed that banks exists in only 53 percent of fishing villages.

25. Tamil Nadu Fisheries Policy (1997) indicated that the performance of the fishermen co-operative societies in the state is generally poor and to a majority of the fishermen co-operative societies are only a means to channel government credit and for other purposes thy are as good as non-existent. Reliable sources indicate that membership in co-operative societies in Tamil Nadu is sought mainly to facilitate access to the welfare schemes run by the state (GAIS, SCR and housing). It is also reported that repayment of loans taken in general has been poor.

26. Bhavani (1998) in her study suggested and concluded that there is very little information available on the nutritional status of fisherfolk communities, due to the fairly low number of baseline surveys conducted on the living conditions of fisherfolk.

27. SIFFS (1998) conducted a survey in Trivandrum district in the year 1996-97 found that about 40% of the fisherfolk houses were not electrified, in a state boasts of 100 percent electrification of its villages.
28. Chennubhotla et al. (1999) conducted a study about the different kinds of non-mechanized and mechanized craft used and gear employed along the Andhra Pradesh coast. They have observed that until the middle of 1960s, fishing for marine finfishes and shellfishes along the Andhra Pradesh coast used to be carried out employing indigenous non-mechanized crafts. Subsequently, trawlers and later mechanized vessels operating gillnets came into use which resulted increased fish production. Use of outboard engine on indigenous crafts for reaching fishing grounds is a recent feature of near the coast fishing.
29. Nayar and Gupta (1999) concluded that in developing countries such as India, marine fish production is far lower than the estimated capacity, currently, fishing activities are carried out mainly in coastal waters; fishing areas in high seas and outside of the country's EEZ are under exploited. Methods of fishery management are ineffective and advanced technologies for fish detection are practically absent. With the launching of the Indian Remote sensing satellite, it is belived that India will be in a position to fully exploit its resources.
30. Yohannan et al. (1999) in their study pointed crafts has meant a larger share of the common wealth of marine fishery. About, 50% of the fish output cornered by the large-scale sector and another 40% by operators of large boats in the motorized sector and the traditional fishermen, especially those in the non-motorized sector, find themselves to be marginalized. He also concluded that the artisanal fisherfolk responded to the new development by going in for motorizing their country crafts. The expansion of motorization was very fast, and in order to cope with incentive compentition new types of gears (nets) like ring seines were also introduced.

31. The Eighth Five Year Plan (1992-97) pointed out the emphasis on the motorization of traditional fishermen income and thereby uplift their standard of living.
32. SIFFS (1999) is a non-government apex organization of villages and district level fish marketing societies of small scale artisanal fisherfolk of south India. The organization has made important contributions in a wide ranger of areas, relevant from a socio-economic prospective, credit and saving for small scale fishermen. It also engages in various development and commercial activities. Commercial activities which include boat building, import and distribution of outboard motors (OBM) and spares, a network of outboard motor workshop and service centres, fisherfolk credit system, growing albyzzia timber for catamarans (the traditional fishing craft) and manufacture and distribution of ice boxes.
33. Thadeus (1999) suggested that fishery community is the society of those who are part of the ancestry and culture of fishing, and belongs to the castes that follow fishing as the traditional occupation. In this study he gives a detail account of the different sections and sub sections of the community and the social status of fisherfolk.
34. Rajn (2000) in his study the present socio economic conditions of traditional fishermen community by examining the effect of MATSYAFED in the traditional economy of the state.
35. Reuben et al. (2000) in their study stated that fishing community is the poorest group of people in all sectors of the economy. Majority of the people look weak, poorly dressed, drunk and live in poor housing structure, and are economically backward. They have a large household size to maintain but their income is not enough to meet their expenditure.
36. Observer (2000) in his study pointed out that most of the fisherfolk among 17 coastal districts borrows money from the local money lender in absence of loan facilities from the scheduled banks. Absence of collateral, to small loans, poor

communication network, and poor payment rate are the major constrains for credit from the scheduled bamks.

37. Fisheries Department (2000) reported that there are 56,169 members in 245 inland fishermen cooperative societies in Tamil Nadu. Most of them are living below poverty line.

38. SDMRI (2000) analysed the implemented activities to make coastal communities in five selected villages along the Thoothukudi coast by providing opportunities for income other than fishing to improve the livelihood of fisherfolk. They have been made to raise the awareness of increasing problem among fisherfolk

39. ICM (2000) revealed that the number of boat owning fishers has declined, and many of the earlier owners have started to work as crew on motorized boats. At least a quarter of the total fishing boats were reported to be owned by outside traders.

40. Government of Kerala (2000) analysed the several interesting initiatives for education among fishing communities. Fisheries Technical High Schools were started in 1968 to enable fishermen with secondary education to learn modern techniques of fishing. Apart from this, residential fisheries schools were started in several districts, where boys from bonofide fishermen families who completed their primary and upper primary education were admitted in class 8 and taught until class 10. These schools provide a public-school type of education with a fisheries bias. Children are also given instruction in seamanship and navigation and practical training in net fabrication. In 1984-85, the state Government started vocational higher secondary courses in fisheries in selected Fisheries Technical High Scholl and Government High Schools with emphasis on fishing craft and gear technology. Fishermen training centers were also started by the state to enable the traditional fishermen to get used to mechanized fishing operations.

41. Kurien and Paul (2000) in their study analyzed the innovative schemes for fisherflok welfare scheme that aims to induct workers engaged in the fishery-related activities as members of the welfare fund. There are nine schemes for allied workers including Group Accident Insurance Scheme, Old Age Pension, SSLC cash award and scholarship, financial assistance for treatment of fatal diseases of allied workers, financial assistance for the marriage of daughters of fisherfolk. In terms of range and coverage, the schemes are really impressive.
42. Rajan (2000) in his study concluded that the traditional fisherfolk were undertaking fishing with a primarily subsistence focus with a sense of camaraderie and community participation. Through continuous interaction with the ocean and fish, the artisanal fisher folk have accumulated trans generationally a treasure of scientific knowledge on diverse marine eco-system and fish behaviour. The new modes of fish production and distribution have resulted in loss of traditional skills and knowledge system, and have converted fishing passive gear to an active gear technology; a low cost to a high cost technology; and from an eco-destructive technology.
43. NETFISH (2000) reported that training programmes were concluded in different coastal states of India to improve the livelihood status of fisherfolk and high fish catching. Altogether 17 training programmes were arranged in Tamil Nadu. At Thoothkudi region, 6 extension programmes on fish quality management were conducted for the fisherfolk in and around Thothukudi fishing harbour.
44. Zein (2000) in his study revealed that food consumption is an important household expenditure of fisherfolk. As the fishermen group is in the lower income group, it is very difficult to fulfil the family nutrition requirement. Their ability to obtain the source of food which have protein and energy is very low. With low earnings, the orientation of family instructed for the requirement of food especially in accomplishment of carbohydrate for all the family.
45. DPHE- DANIDA (2000) reported that fisherfolk in coastal zone generally use tube well water for drinking purpose but for other household activities,

including household washing, bathing and cooking they use pond and river water. He also stated that 60% of the fisherfolk defecated in fixed place (pit latrine) but most of the latrines are not up to the mark from sanitary point of view. As a result, diarrheal diseases and parasitic infections are common within the coastal community, especially the coastal community, especially the fisher folk.

46. FAO (2001) reported that some anecdotal evidence exists to show that fishing communities have above average poverty rates, but few hard data and analyses are available on the nature and extent of poverty in their communities and on the relative importance of different causes of poverty and on the most effective actions to alleviate poverty.

47. Government of India (2001) pointed out the objective of saving-cum-relief (SCR) is to provide financial assistance to the fisherfolk during lean season. The rate of contribution by the fisherfolk is Rs.75 per month for eight months. This contribution is matched by a contribution shared equally by the central and state governments, and the accumulated amount is distributed back to fishermen in four equal instalments of Rs.300 per month.

48. Duraiappah (2001) in his study indicated that the coastal communities use natural resources primarily as an asset for income generation. It is recognized that environmental resource also provide life supporting services and confer many intangible aesthetic and cultural benefits.

49. Government of Delhi (2002) in their study analysed the poverty line of fisherfolk as the cost of an Indian average consumption basket at which calorie norms was met. The norms were 2400 calories per capital per day for rural coastal areas and 2 100 calories for urban areas. The poverty line serves as a cut-off line for separating the poor from the non-poor. Survey data on household consumption expenditure collected by the national sample survey organization (NSSO) were used for the purpose.

50. Annabelle Cruz et al. (2002) in their study pointed out that the socio-economic condition of the rural fisherfolk is in a backward condition. As the house hold size of the fisherfolk family is not enough to meet their expenditure. In such situations they borrow money from money lenders. They lack all the basic amenities which are important for a quality life.

CHAPTER 3

AN OVER VIEW

3.1 FISH AND FISHERIES: AN OVER VIEW

India is the third largest fish producing county and the second largest aquaculture fish production in the world. India contributes about 7% to the global fish production. The country is also home to more than 10% of the global fish biodiversity and is one of the 17-mega biodiversity rich countries. Around 14 million people are engaged in fisheries and its allied activities. Andhra Pradesh is the largest fish producer in the country followed by West Bengal and Gujarat. The total fish production during 2017-18 is estimated to be 12.60 millions metric tonnes, of which nearly 70% is from inland sector and about 50%of the total production is from culture fisheries. More than 50 different types of fish and shellfish products are being exported to 75 countries around the world. Fish and fish products have presently emerged as the largest group in agricultural exports Rs.45, 106.89 cores in value. This accounts for around 10% of the total exports and nearly 20% of the agricultural exports, and contribute to about 0.91% of the GDP and 5.23% to the Agricultural GVA of the county.

Fisheries

Fisheries is an economic activity that involves harvesting fish or any aquatic organism from the wild (capture fisheries) or raising them in confinement (culture fisheries/Aquaculture). It may be Traditional/Small Scale Fisheries (SSF) for sustenance, or Large-Scale/Commercial Fisheries for profit.

Fish

Fish is a cold-blooded aquatic organism that breathes with gills and swims with fins; they are categorized as Finfish and Shellfish.

Finfish

Finfish are cold-blooded aquatic vertebrates that have gills, fins with rays, and scales coverings the body.

Shellfish

Shellfish are cold-blooded aquatic invertebrate that have gills, various types of locomotory organs and a shell/exoskeleton covering the body. They include crustaceans and mollusc.

Biodiversity

India has a large number of finfish species. As per the database of the National Bureau of Fish Genetic Resource (NBFGR), Lucknow, 2,508 species of native finfish have been recorded, of which 1,518 species are from the marine environment, 113 from brackish waters and 877 are from freshwater habitats. In addition, 291 exotic fish species fish species also occur in India.

Table: 3.1

Fish Diversity of India

Native Fishes	**Number of Species**
Marine Ecosystem	1518
Brackish water Ecosystem	113
Fresh water Ecosystem	877
Sub-total	**2508**
Exotic Fishes	**291**
Total	**2799**

Source: Secondary data

Categorization of Fish by their habitat:

- **Freshwater Fish**: Fish that spend most or all of their life in freshwater, such as rivers and lakes, having a salinity of less than 0.5 ppt. Around 40% of all known species of fish are found in freshwater.

  They may be divided into **Coldwater Fish** (5-20 ^{0}c); examples: Mahseer, Trout, etc., and **Warmwater** Fish (25-35 ^{0}c); example: carps, catfish, snakeheads, featherbacks, etc.
- **Brackishwater Fish**: Fish that can tolerate a wide range of salinity (0.5-30.0 ppt) and live in backwaters, estuaries and coastal waters.

  Example: Mullet, Milkfish, Seabass, Pearlspot, Mudskipper, etc.
- **Marine Fish**: Fish that spend most or al of their life in seawater, such as seas and Oceans, having salinity above 30 ppt. There are about 240 species contributing to the marine fisheries.

  Examples: Sardines, Mackerel, Ribbonfish, Anchovies, Grouper, Cobia, Tuna, etc.

3.2 FISH STOCK DEPLETION IN INDIAN INLAND OPEN WATERS

Indian fisheries resources have been playing an extremely important role in providing livelihood supports and employment opportunities to millions of fisher-folk and fisheries stockholders. Indian inland fisheries are endowed with diverse and large water resource in terms of ponds, tanks, floodplain wetlands, lakes, reservoirs, rivers, which has huge fishery potential contributing significant export earnings and thus national development. All these waterbodies provide a unique habitat to support various flora and fauna including fish and shellfish. Fish has turned into an essential constituents in the food spectrum of most Indians which provides healthy nutrients in terms of high quality protein, lipids, vitamins and minerals. Although fishes of large-size are

costly and cannot be afforded by common people on regular basis, small-sized indigenous fishes could be considered a poor man's protein because these can be easily afforded by economically poor people assuring food nutritional security. About 60% of Indian population consumes fish and the apparent per capital fish consumption was estimated to bear around 10 kg (NFDB, 2018). Fishery resources of India are divided into two categories: marine fisheries and inland (freshwater) fisheries. Inland fisheries are classified as freshwater aquaculture, including the pond culture of carp; brackishwater aquaculture, involving mostly shrimp culture; and capture fisheries in rivers, estuaries, lakes, reservoirs, etc. The freshwater aquaculture production in India comprises about 2.36 million ha of pound and tanks which accounts for nearly 55% of the total fish production in India (NFDB, 2018). The increasing trend in inland fish production is mainly due to application of advanced package-of-practices and technologies in aquaculture. Off late, the inland fisheries sector is faced with tremendous anthropogenic pressure leading to loss of habitat and biodiversity, and altered nutrient cycling causing decline in fish production. Climate change is also contributing to the problem in inland fisheries.

MAJOR THREATS TO INLAND OPEN WATER BODIES

India is one of the mega biodiversity nations which occupies ninth position in terms of freshwater diversity and having four biological diversity hotspots with diverse fish germplasm resource. Overall, 3055 fish species are recorded from India that contributes approximately 11.72% of fish species recorded worldwide (NBFGR 2018). Unfortunately, these rich fish germplasm resource are under severe threat from various natural as well as anthropogenic activities due to rapid and uncontrolled industrialization, urbanization and other related

infrastructural development activities. Some of the causes of fish stock depletion are industrial, domestic and pesticides pollution, habitat destruction, land development, soil erosion and overexploitation of fish. In this scenario, it is required to take responsibility to create awareness among the masses about declining fish catch in inland waterbodies, which can contribute to long-term sustainable utilization, conservation and management of the open water resources. Some of the important causes of depletion of fish fauna are discussed.

ANTHROPOGENIC CAUSE

The major anthropological stresses are widespread water pollution due to effluent discharge from industries, domestic wastes such as detergents, pesticides etc. Other observed changes are indiscriminate fishing, use of non-selective gear, river embankments, sand digging and hydropower. Introduction of exotic species, intentional or unintentional, in inland water is affecting native fish fauna.

OVEREXPLOITATION

Overexploitation (over-fishing) of fish stock is one of the major threats to the aquatic biodiversity which affects livelihoods of fisher-folk (FAO, 2018). Over the last decade, the harvest from inland capture fisheries has stagnated. Overfishing is one of the major reasons for the same. There are two types of overfishing, through the destruction of broodstock (gravid stock) called recruitment overfishing and juveniles fishing called growth overfishing. Overfishing reduces population size. Depending on habitat, overfishing affects feeding and breeding grounds of fish funa; indiscriminate fishing of brood fish & juvenile, trophic level change and ecosystem damage consequent to overfishing can lead to fish proteins, market pressure, application of large-scale fishing gear technology, lack of appropriate management approaches and

policies, large volumes of by-catch are the major concerns for sustainable inland fisheries.

AQUATIC POLLUTION

Pollution is one of the most significant and noticeable factors responsible for decline of populations of many fish species in rivers and lakes. Industrial wastes, sewage disposal, pesticides and insecticides used in agriculture field and improper use of wastewater treatment plants result in an increase in the discharge of emerging pollutants into the aquatic environment. All these wastes usually carry non-biodegradable constituents, which result in decreased oxygen levels and cause mass mortality of fishes at times. Pesticides, industrial effluents, untreated or semi-digested sewage often lead to excessive nutrient enrichment from non-point source in rivers, lakes and estuaries in India causing increased biological oxygen demand and depletion in dissolved oxygen levels of those waterbodies.

HABITAT DEGRADATION

Habitat degradation is caused by a series of interactive factors bringing in changes within the drainage basin. Rapidly increasing deforestation to a accommodate abruptly growing human population and their present needs, the problem of soil erosion and their present needs, the problem of soil erosion and siltation is increasing; as a result water level is decreasing, which adversely affect the fish population through loss of habitat, breeding, nursery and feeding grounds and there by results in stock depletion of many fish species. Reduction or even loss of floodplains by damming, agriculture practices, urban development, rivers dredging and geomorphological modifications may lead to the decline of our endemic and indigenous species.

INTRODUCTION OF EXOTIC SPECIES AND ITS NEGATIVE IMPACT

Introduction of exotic or "alien" invasive species in freshwater ecosystem is believed to be one of the greatest threats to native species. Due to their rapid growth compared to native species, it adversely affects their stock; exotics normally grow faster through competing for space and food. Lack of attention to identify and to develop advanced aquaculture technique for suitable local species appears to have resulted in the dominance of exotics and their impacts. Exotic species can impact biodiversity either directly or indirectly and these impacts can be immediate or long-term. Direct impacts include predation, disease outbreak, endangering them gradually leading to extinction. Indirect impact includes ecological, economical and genetical and also include habitat destruction. The potential impact of exotic species on biodiversity cannot be ignored easily because high-impact invaders are more likely to belong to genera not already present in the system.

CONSTRUCTION OF DAMS

Construction of dams, embankments, antiques and barrages adversely affects the feeding and breeding ground and migration routes of many commercially important fishes such as hilsa and mahseer and also small fishes. Failure of stock recruitment process takes place due to the destruction of breeding and spawning activities. Construction of Farakka barrage on river Ganga has drastically affected the abundance of Tenualosa ilisha. A number of dams and barrages have been built in the upper stretch of river Ganga and Brahmaputra and these have been the major causes for the reduction of natural riverine water flow leading to obstruction of feeding, spawning and migration routes of mahseers, snow trouts etc. Have been blocked responsible for the decline of these fish stocks.

FORESTRY PRACTICE

Due to afforestation and deforestation, freshwater habitats and fishes are subjected to serious threats. The effects of each stage of the forestry cycle from ground preparation to the maturing crop and felling possess a negative impact on local freshwater. Forestry practices can lead to alteration in physical and chemical characteristics of water temperatures and extreme fluctuations in water temperatures and extreme fluctuations in water chemistry. Inland water channels can have reduced water temperature in areas shaded under afforestation and higher temperature near the catchment that has been deforested. Increased nutrients alter plant communities and cause algal bloom formation in streams and lakes. Due to debris in deforested streams, fish food availability is reduced in river channels. Acidification of water also affects the composition of flora and fauna, eliminating fishes in some cases.

LACK OF LOW IMPLEMENTATION AND AWARENESS AMONG MASSES

Lack of proper implementation of laws and guidelines for inland fisheries resulted in many problems facing the sector. There is no proper monitoring and limits on daily fish catch, size of fish caught, and mesh size of nets used, which affects, juvenile fishes and the topic ecosystem. Adding to the problem is lack of awareness. Poor implementation and adoption of fisheries legislation, conservation and sustainable measures result in fishing in unplanned manner. Hence, proper enactment of guidelines and laws, and training the local people on sustainable use of waterbodies are required for conservation.

PROBLEMS IN DATA REPORTING OF INLAND FISH CATCH

There is lack in scientific or authentic information or data about fisheries in lentic and lotic water systems of India. It severely affects the decision making process. Lack of accurate and authentic information renders, analysis of existing data sets difficult and makes decision –making problematic.

Thus, it is of prime importance to maintain data related to stock, yield etc. Collection of inland fisheries in India is a small-scale activity in nature, highly dispersed and generally not reported accurately to government agencies and also there are no direct estimates of fish harvest as many of the fish caused in those places are either consumed directly or sold in local markets by artisanal fishers. However, attempts should be made to estimate inland fish catch by artisanal fishers.

NATURAL CAUSE

CLIMATE CHANGE AND ITS IMPOACTS ON FISH DIVERSITY AND ECOSYSTEM

Due to the rapid increase of greenhouse gas emission in the atmosphere because of increasing burning of fossil fuel global climate has been changing rapidly since last few decades and posing greatest threats to sustainability of fisheries. It has great impact on inland fisheries and several studies found that the catches of most demanded Indian major craps and catfishes has been declining day by day from the riverine and wetlands resources due to irregular seasonal rainfall, siltation, prolonged drought. The monsoons have become quite irregular; even when they arrive on time, they seems less consistent in their behaviour. Changing seasonal pattern of rainfall affects fish catches. The river and streams water flow can influence water chemistry and population dynamics. Changing temperature regimes in waterbodies affects availability of fish food organisms. Climate change is likely to produce profound modification to the structure and functioning of the aquatic ecosystem potentially changing the distribution and abundance of many species.

DISEASE OUTBREAK

Among various diseases, the bacterial, fungal, protozoan parasite and helminth are common in inland open water bodies and threatening many species

of fish. Inflation of Aphanomyces invadans that causes epizootic ulcerative syndrome is still posing serious threats to number of populations of commercially important indigenous species in major part of the country especially during winder season.

SILTATION AND WEED INFESTATION

Siltation is a major problem observed in wetlands of India due to loss of connectivity with the parent river or abstraction of water for irrigation, pollution, unusual rainfall, prolonged drought and excessive runoff from the vicinity of agriculture fields. Siltation from catchment area has been obstructing the feeding and breeding and nursery grounds of many commercially important as well ecologically important fish stocks. It affects the wetland by raising the waterbed and consequently reduces the water depth in the wetlands, which results in loss of riverine connectivity and subsequently loss of fish stocks. Some of the wetlands in the country cannot retain water for more than two to three months of the year. Further, siltation has facilitation encroachment. Weed infestation has also been affecting fish production and fish recovery in many wetlands of North-eastern part of India. The decomposition of macrophytic weed and its silt load reduces the water depth of inlet/outlet channels has reduced the auto-stocking capacity of the wetlands. In some of the wetlands, fishermen have been responsible for the destruction of inlet/outlet channels with the construction of bundhs for various purposes.

EXTREME WEATHER EVENT

Increasing frequency and intensity of extreme weather events such as drought, unusual flood occurrence, landslide, river erosion etc. are becoming a major factor for declining fish diversity, loss of fish stocks, habitat loss and degradation in inland open waterbodies.

3.3 DWINDLING FISH STOCK: KERALA PLANS TO REGULATE MECHANISED FISHING

With the rapid depletion of fish stocks in Indian waters causing concern, the state Fisheries department has decided to regulate fishing activities along Kerala's coastline.

Though Kerala was the leading state in marine fish production in India, it slipped to fourth position during the last decade due to over exploitation of fishing resources.

Fisheries minister J Mercikutty Amma said that the government was actively considering amendments to Kerala Maine Fishing Regulation Act, 1980 to control the wanton use of deep sea trawlers, which is threatening the livelihood of traditional fishermen. More regulations will be incorporated in consultation with fishermen. The government is planning to ensure sustainable fishing by regulating mechanised fishing.

Under Kerala Marine Fishing Regulation Act, 1980 (KMFRA) only the nets used in trawling boats are regulated. Standardisation of mesh size was a long over due process. Rampant netting of fish fry using nets of narrower mesh size has to depletion of fish stocks.

Tonnes of yough fish are netted and sold to the manure industry. Even fries are netted and sold to the manure industry. Even fries of highly valuable fish varieties such as seer fish are caught. This unethical practice has led to steep decline in fish population along Kerala's coastline.

Considering the ecological consequences of overfishing the government had appointed many commissions to recommend steps to check unscientific and improper fishing practices. The commissions recommended ban on fishing nets with a mesh size below 28 mm including Chinese and trammels nets. "The

commissions recommended ban on fishing practices. The commissions recommended ban on fishing nets with a mesh size of 33 mm.” said a scientist from Central Marine Fisheries Research Institute (CMFRI).

The state government is also planning to reduce the number of fishing boats operating in the state. “Issuing fishing licence to boats as well as boat yards is under consideration. Some of the radical measures such as the Norway model of fixing quota for catch are in the discussion stage.” Said a senior a senior official with Fishing using purse seine nets, use of Chinese nets during high tide, various trawling practices such as pair trawling, bull trawling would be banned.

CHAPTER 4

DATA ANALYSIS AND INTERPRETATION

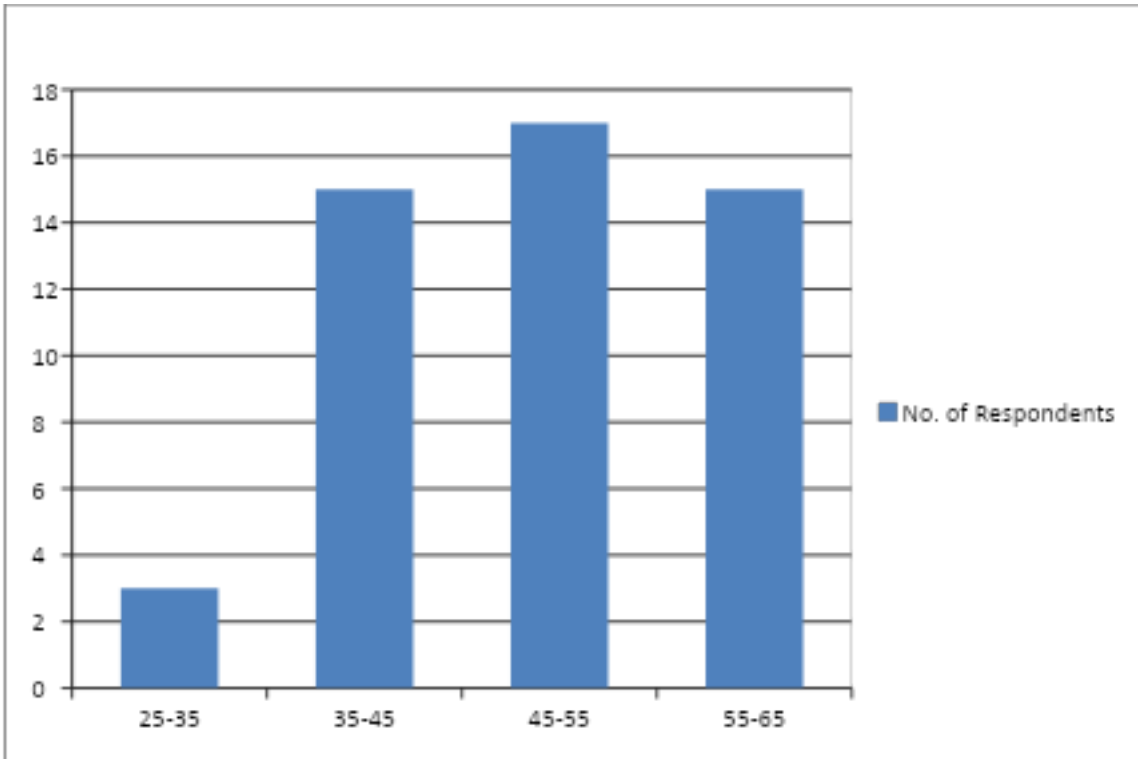
4.1 Age wise distribution of fishermen

This table shows age wise classification of the respondents. It reveals the fact that from 50 respondents, only 6% belongs to age 25-35, 30% belongs to 35-45, 34% belongs to age 45-55 and 30% belongs to above 55 years.

Table: 4.1

Age group	No. of Respondents	Percentage
25-35	3	6%
35-45	15	30%
45-55	17	34%
55-65	15	30%
Total	50	100%

Figure:4.1



Source: Primary data

When the age structure of the respondents is observed a notable fact is that compared to other age group more than fishermen working under the 45-55 categories. Therefore, it is clear from this data that this is the time when the family needs more economic support. The another fact is that only 6% working under the 25-35 category. The age classification reveals that majority of fisherman are aged and experienced.

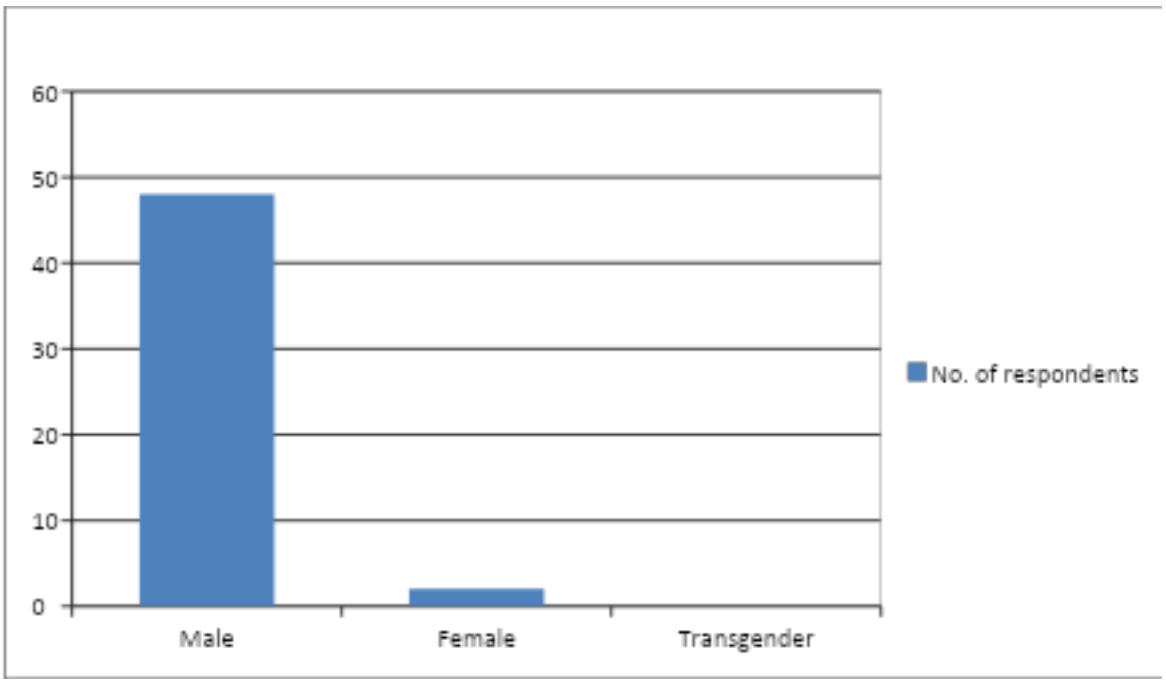
4.2 Gender of Respondents

The following table is discussing about the gender of the respondents.

Table: 4.2

Gender	No. of respondents	percentage
Male	48	96%
Female	2	4%
Transgender	0	0
Total	50	100%

Figure: 4.2



Source: Primary data

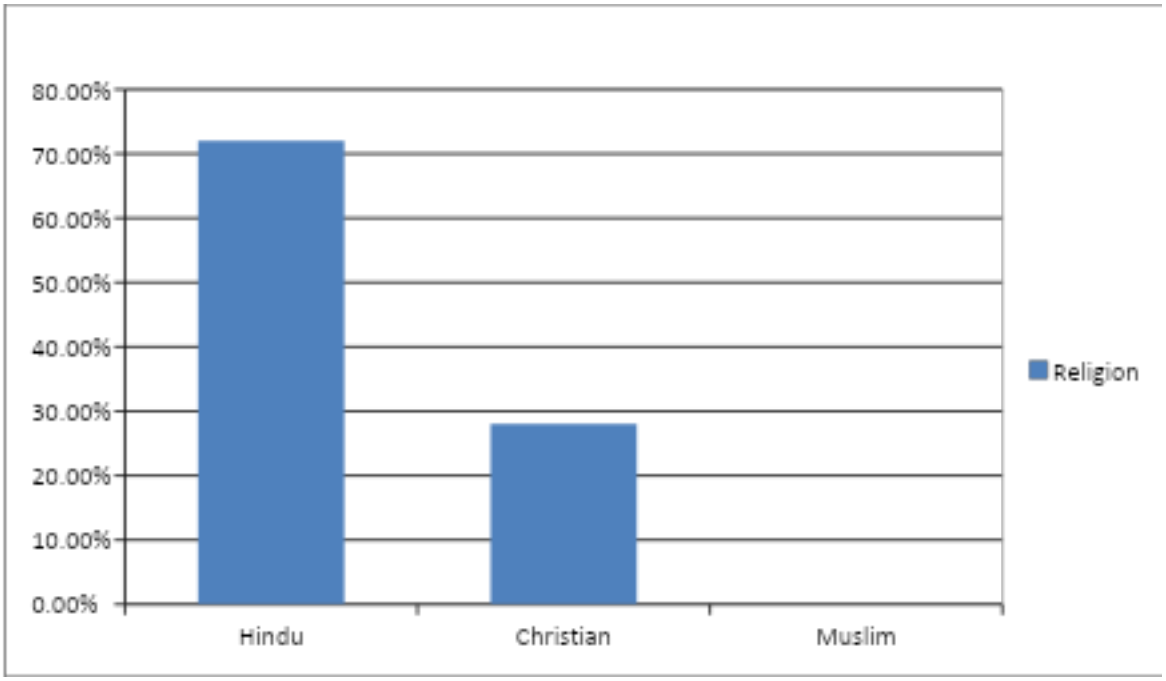
The above table and figure shows the gender of respondents. It shows that 96% of respondents are male and 4% of respondents are female. There is no transgender, respondents involving fishing. Usually, men are going for fishing than women.

4.3 Religion of the respondents

Table: 4.3

Religion	No. of respondents	Percentage
Hindu	36	72%
Christian	14	28%
Muslim	0	0
Total	50	100%

Figure: 4.3



Source: Primary data

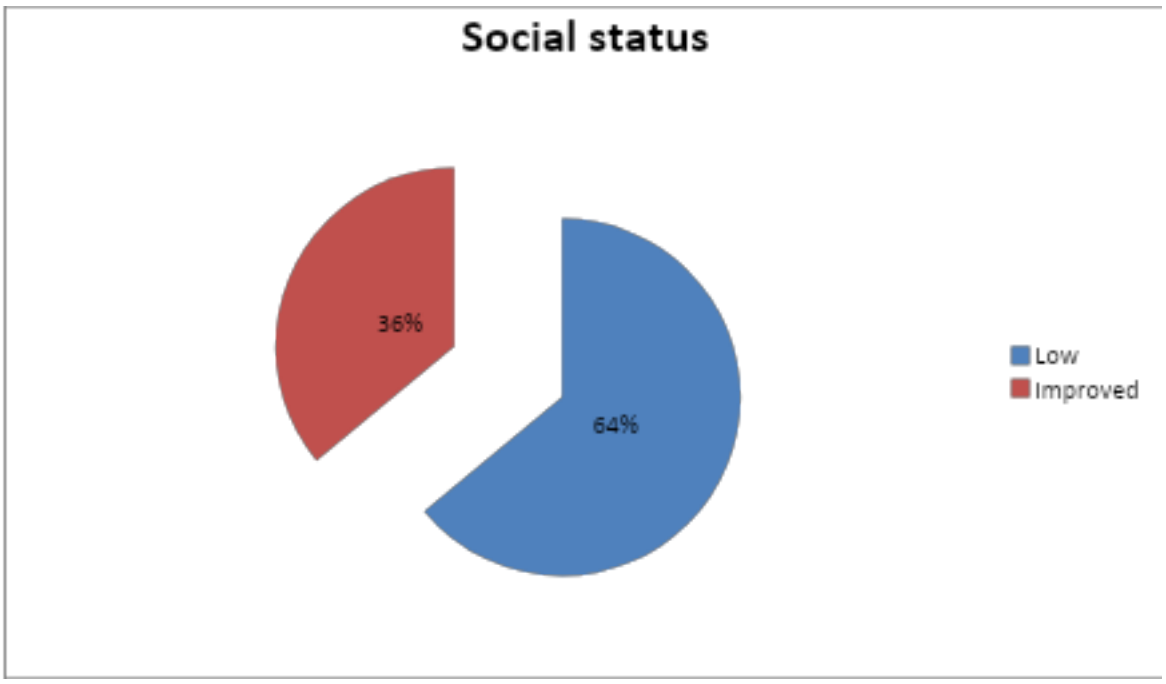
From the table it is clear that the most of the respondents are from the Hindu community. About 70% of the respondents belongs to Hindu Community and 28% of the respondents belongs to Christian and there is no respondent from Muslim community.

4.4 Social Status of the Respondents

Table: 4.4

Social status	SC	ST	OBC	Other	Total	Percentage
Low	7	3	20	2	32	64%
Improved	2	1	6	9	18	36%
Total	9	4	26	11	50	100%

Figure: 4.4



Source: Primary data

This table shows social status of the respondents. This shows the distribution of fishermen according to their caste 52% members are belonging to OBC and 22%of the respondents belong to general categories and 18% members belonging to SC. The ST categories are very less. In the figure shows most of the respondents are belongs to low social status and only 36% of the respondents are improved their status.

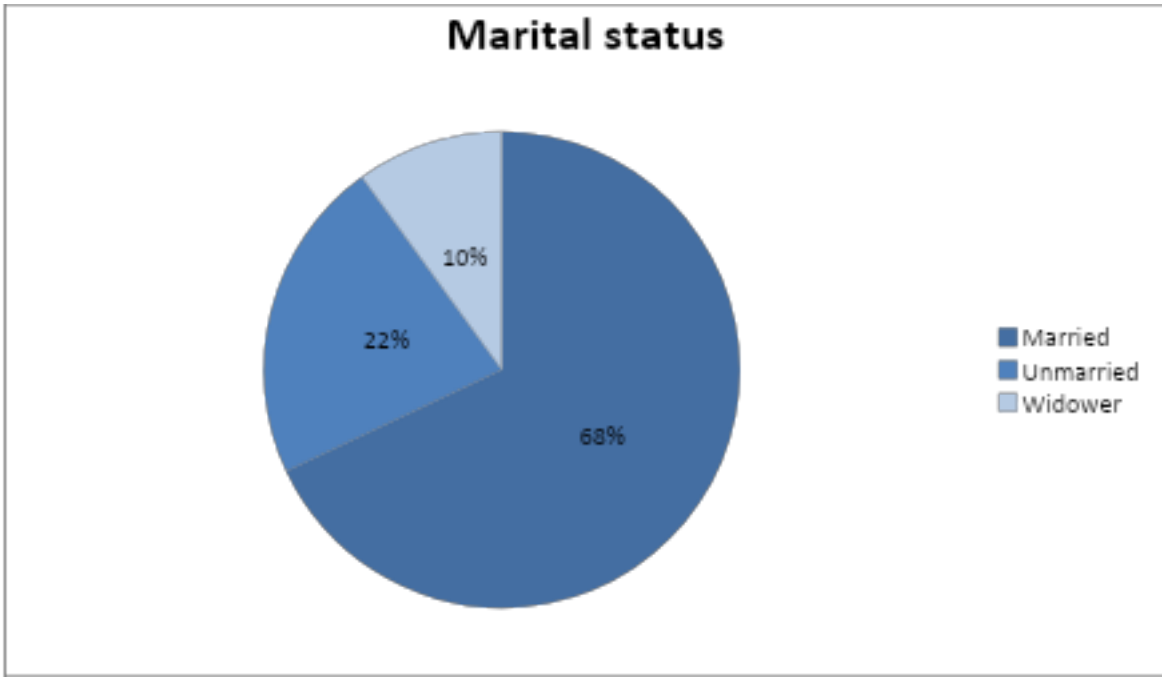
4.5 Marital Status of the Respondents

Table: 4.5

Marital status	No. of respondents	Percentage
Married	34	68%
Unmarried	11	22%
Widower	5	10%

Total	50	100%

Figure: 4.5



Source: Primary data

The above table shows the marital status of the respondents. 22% of the respondents are unmarried and 68% of them are married. 10% of the respondents are widower.

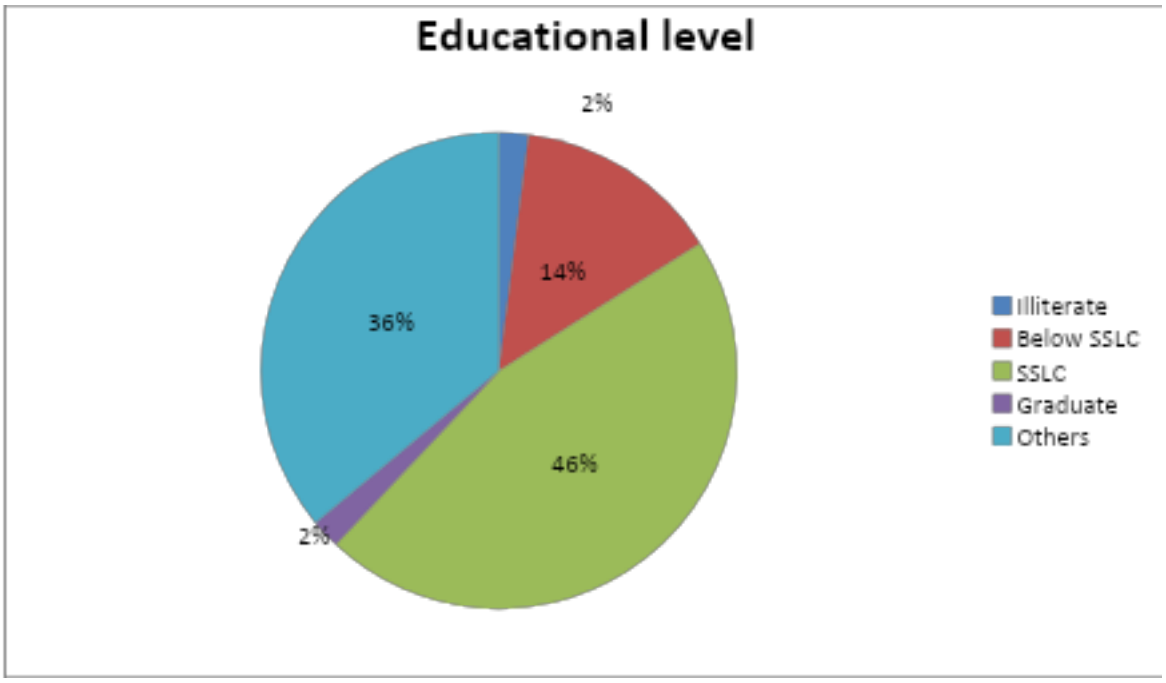
4.6 Educational Background

Education is the major tools for the necessary fulfilment of the individual. Education is the key variable in the exploration of socio-economic status of the individual. Educational levels of the respondents are shown in the table below.

Table: 4.6

Educational level	No. of the respondents	Percentage
Illiterate	1	2%
Below SSLC	7	14%
SSLC	23	46%
Graduate	1	2%
Others	18	36%
Total	50	100%

Figure: 4.6



Source: Primary data

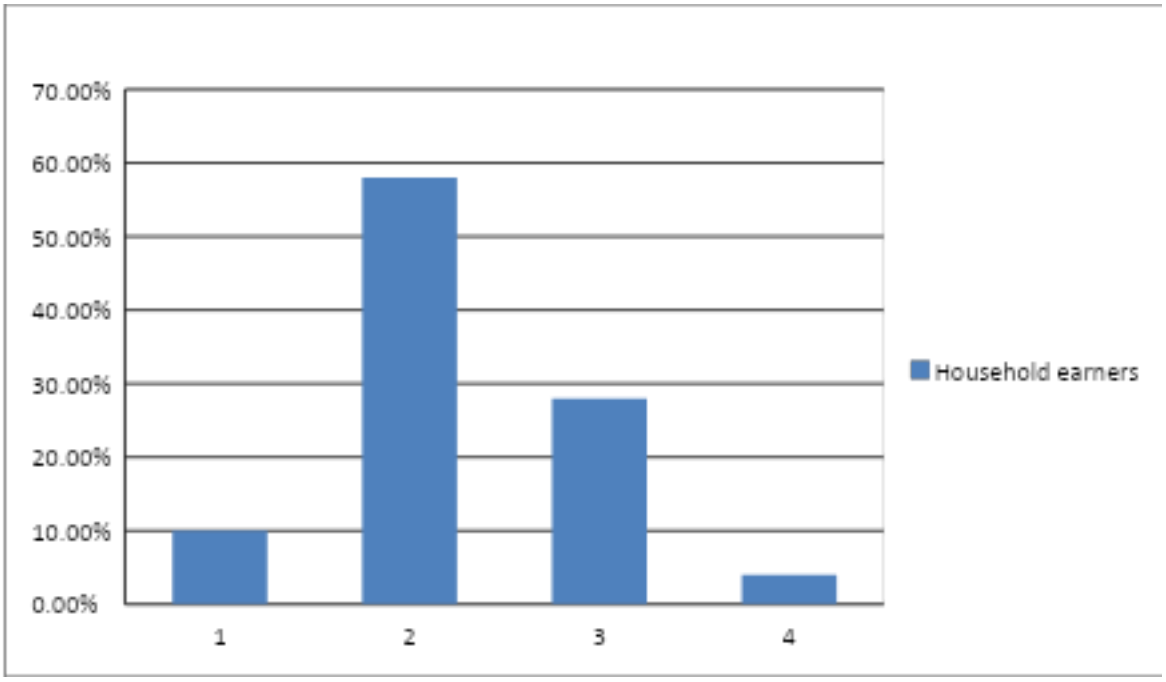
The above figure shows that the majority (46%) of the respondents have been completed SSLC. 2% of the respondents are graduated and 36% of the respondents are other educational level and 14% of the respondents are below SSLC. The figure clearly reveals that most of the respondents are educated and only 2% of them are illiterate.

4.7 Number of Household Earners in Respondents Family

Table: 4.7

No. of household earners	No. of Respondents	Percentage
1	5	10%
2	29	58%
3	14	28%
4	2	4%
Total	50	100%

Figure: 4.7



Source: Primary data

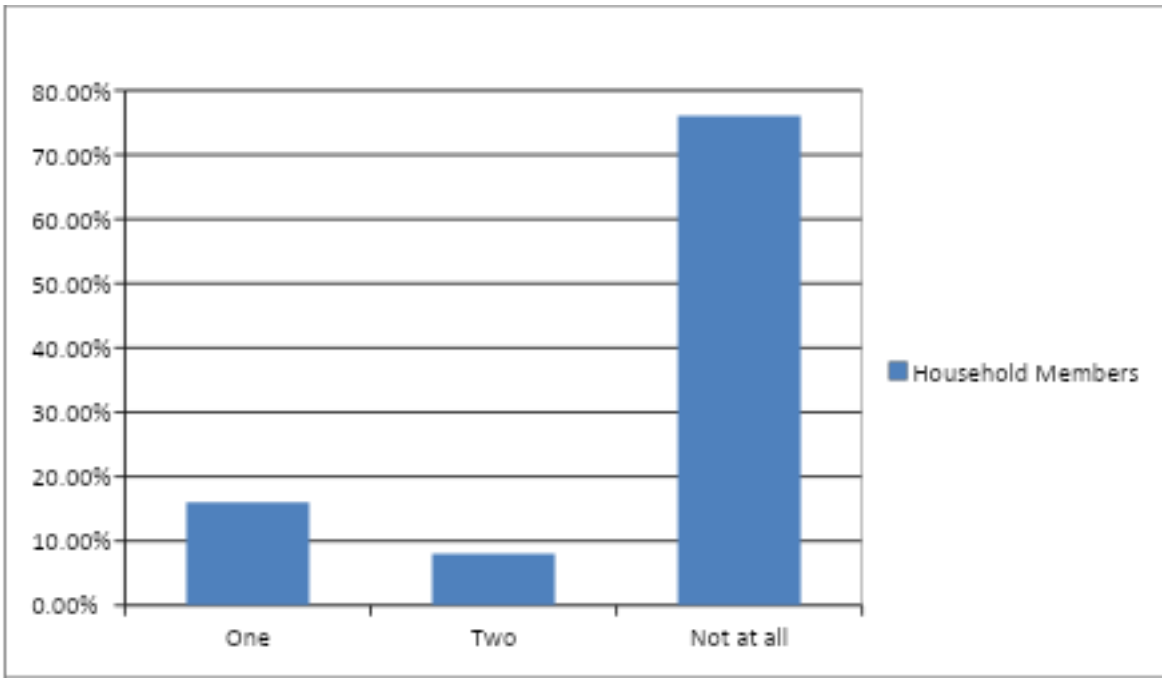
The above figure shows that 58% of the respondent family have two earning members and 28% of the respondent family has three earning members and 10% of the respondent family has only one earning member.

4.8 No. of Household members working with respondent

Table: 4.8

No. of members	No. of respondents	Percentage
One	8	16%
Two	4	8%
Not at all	38	76%
Total	50	100%

Figure: 4.8



Source: Primary data

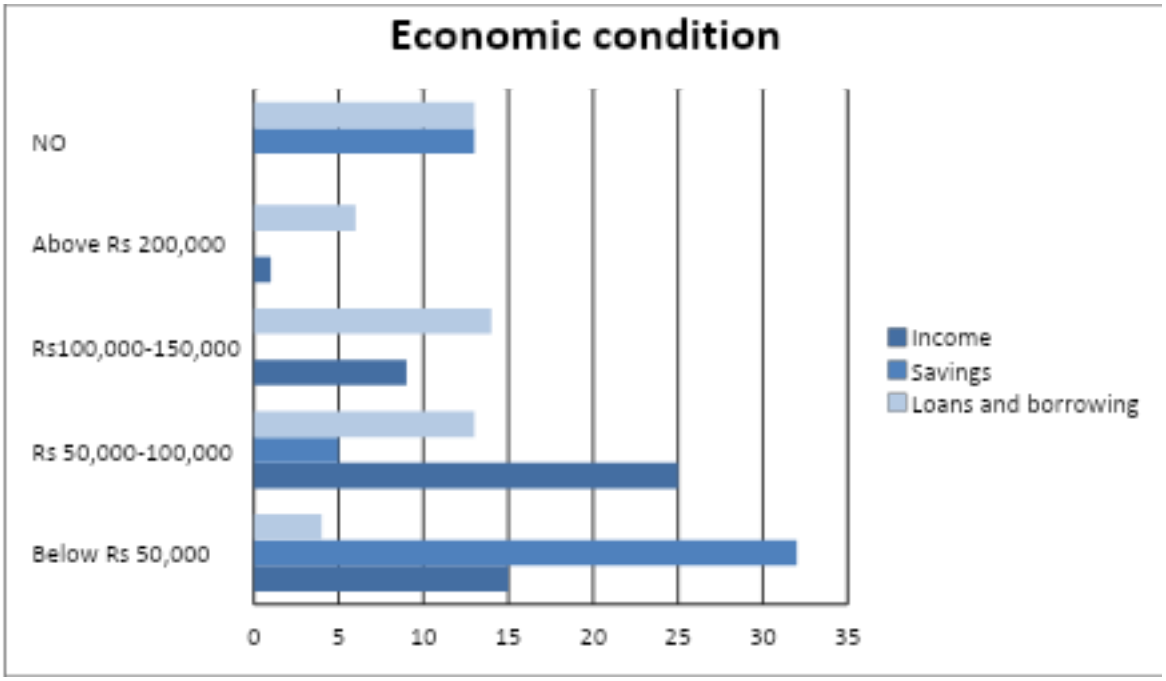
The above figure shows that most of the household members (76%) are not at work with the respondent. Only 8% of the two household members work with respondent and only 16% of the household one member is work with the respondent. It means that most of the household members of the respondent are not willing to work in fishing.

4.9 Income, Savings, loans and borrowings of the Respondents

Table: 4.9

Amount	Income	Savings	Loans and borrowings
Below 50,000	15	32	4
50,000-100,000	25	5	13
100,000-150,000	9	0	14
Above 200,000	1	0	6
NO	0	13	13
Total	50	50	50

Figure: 4.9



Source: Primary data

Income, savings, loans and borrowing wise classification of the sample respondents in the above figure and table shows that out of 50 respondent 30% has annual income below Rs.50000, 50% have income between Rs.50, 000-100,000 and 18% of the respondents annual income between Rs.100, 000-150,000. Only 2% of the respondents have annual income is above

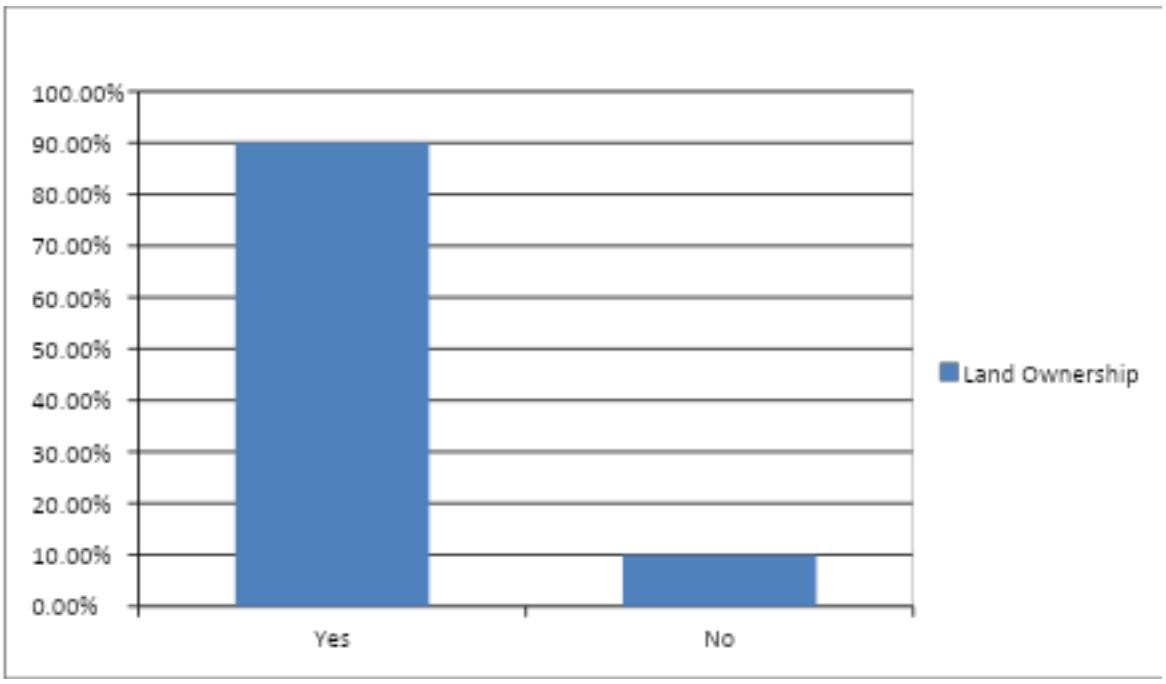
200,000.The another important factor is savings of the respondents that is clear in the figure, 64% of the respondents savings are less than Rs.50,000. Only 10% of the respondents have between 50,000-100,000 savings. 26% of the respondents have no savings. It is clear that 64% of the fishermen they saved little money. Another factor is that loans and borrowings, 28% of the respondents are borrowed Rs.100,000-150,000. 26% of the respondents are to borrow Rs.50000-10000 and 26% of the respondents as no loans are borrowings. In the figure we can understand most of the respondents the level of savings more than loans and borrowings. The reason is the people consuming alcohol. It negatively affects day to day life of the fishermen. Reduce the consumption of alcohol fishermen have been save more money and increase their standard of living.

4.10 land Ownership of the respondent

Table: 4.10

Land ownership	No. of the respondents	Percentage
YES	45	90%
NO	5	10%
Total	50	100%

Figure: 4.10



Source: Primary data

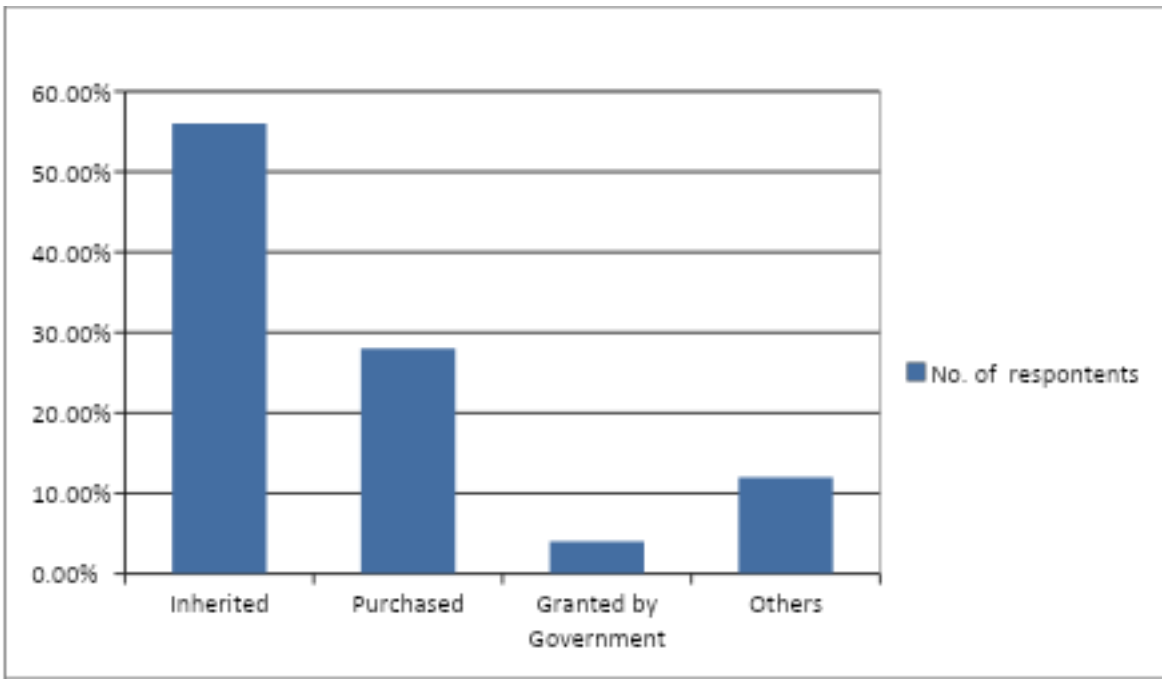
In the above figure shows that most of the fishermen respondents are their own land.90% respondent have land ownership and only 10% of the respondents are landless.

4.11 Type of Land Ownership

Table: 4.11

Type of land ownership	No. of respondents	Percentage
Inherited	28	56%
Purchased	14	28%
Granted by Government	2	4%
Others	6	12%
Total	50	100%

Figure: 4.11



Source: Primary data

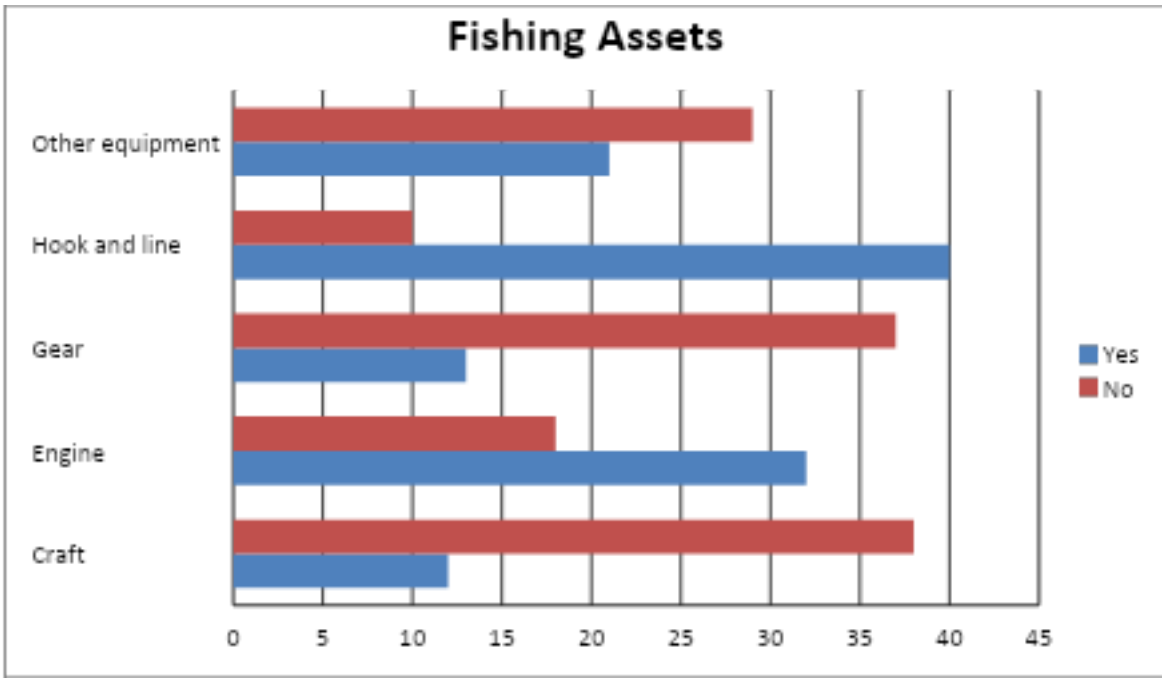
The above figure shows that 56% of the respondents have their own land is inherited. Only 28% of the respondents are purchased by land. Only 4% of the respondents have land given by the government. The remaining 12% are other ownership factors of the fishermen respondents.

4.12 Fishing Assets of the Respondents

Table: 4.12

Fishing Assets	YES	NO	Total
Craft	12	38	50
Engine	32	18	50
Gear	13	37	50
Hook and line	40	10	50
Other equipment	21	29	50

Figure: 4.12



Source: Primary data

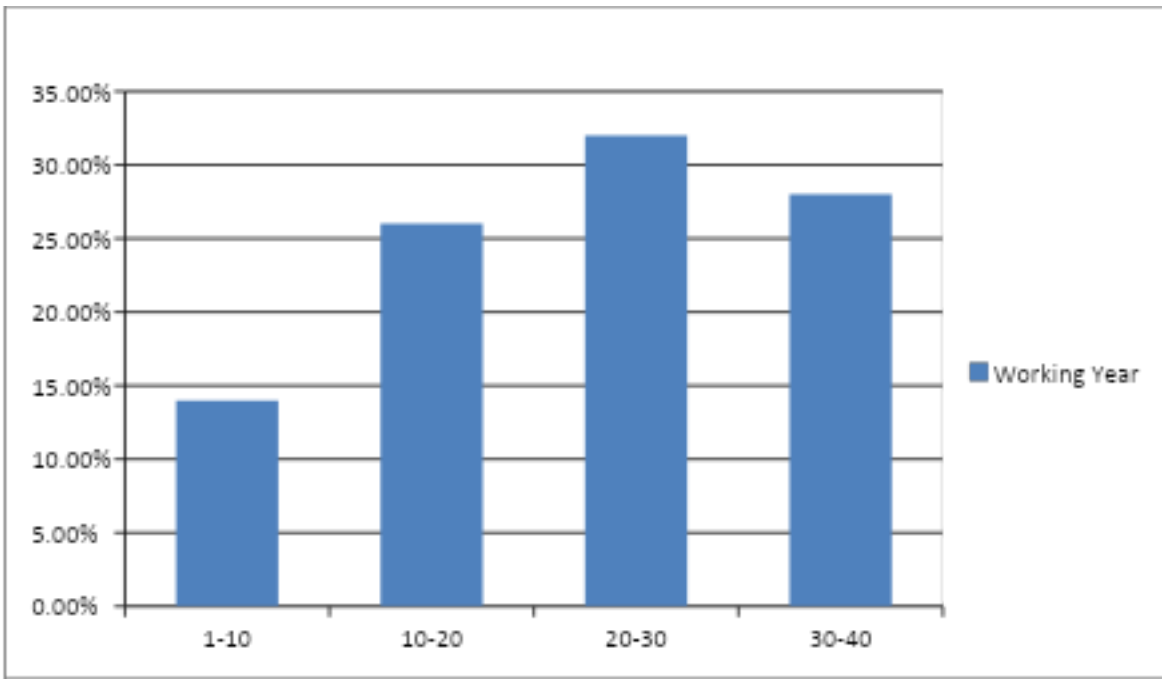
In the figure shows that fishing assets of the fishermen respondents in kumarakom. 64% of the respondents are their own engine for fishing. Most of the respondents have there own hook and line but they are inefficient for fishing. 24% of the respondents have use other equipments in fishing.

4.13 Working year of the Respondents

Table: 4.13

Year	No. of respondents	Percentage
1-10	7	14%
10-20	13	26%
20-30	16	32%
30-40	14	28%
Total	50	100%

Figure: 4.13



Source: Primary data

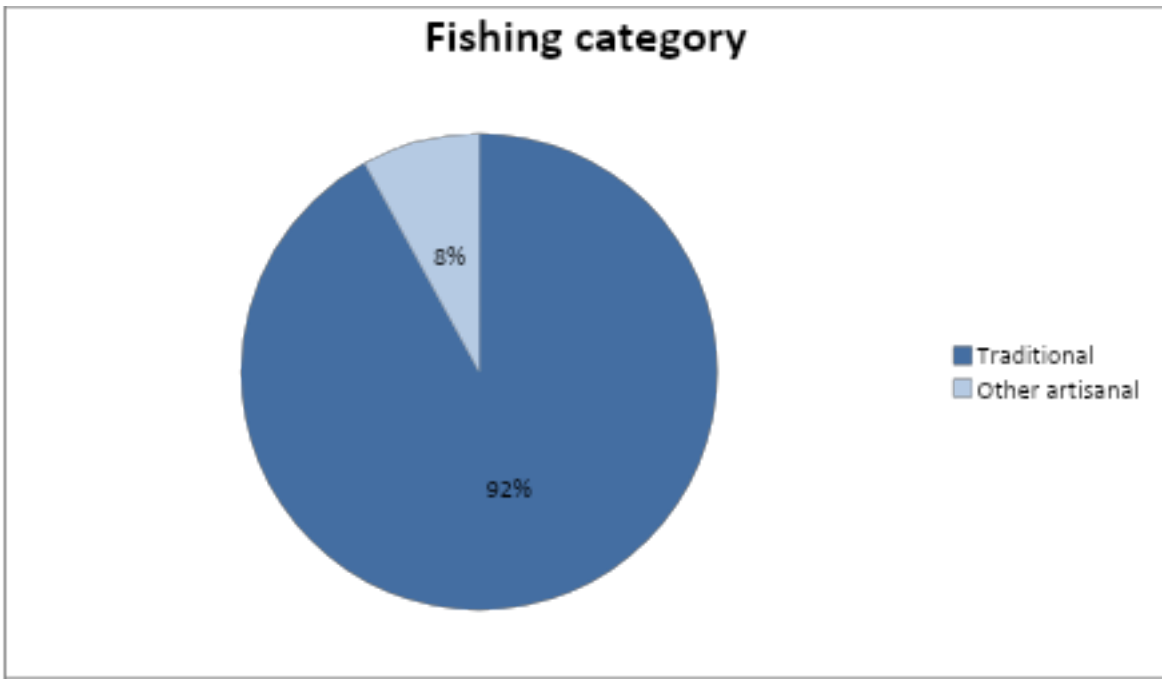
In the figure shows that the working year of the fishermen in kumarakom. In the above table we can find out most of the fishermen in the kumarakom panchayath as well experienced.32% of the fishermen has 20-30 years long experience in fishing.28% of the respondents have 30-40 year long experience in fishing and 26% of the respondents are 10- 20 years experience in fishing. Only 14% of the respondents are in 1-10 years.

4.14 Fishing Category of the Fishermen

Table: 4.14

Fishing category	No. of respondents	Percentage
Traditional	46	92%
Other artisanal	4	8%
Total	50	100%

Figure: 4.14



Source: Primary data

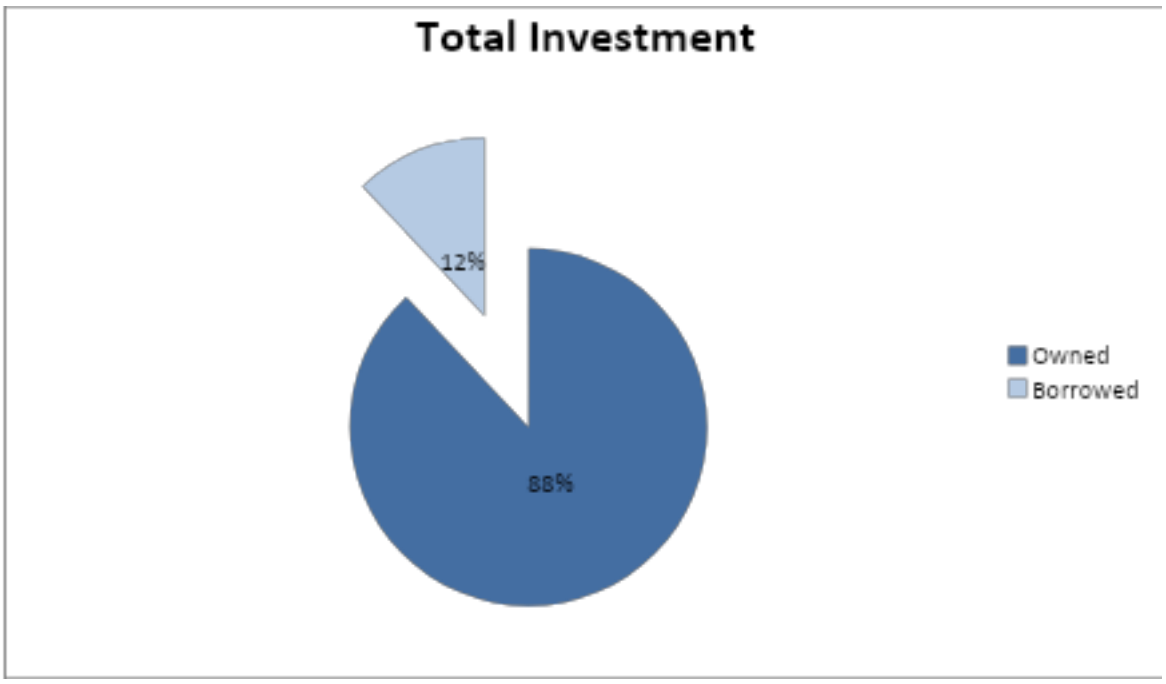
From the above table we can find out fishing category of fishermen in the sample area. A notable fact obtained from the above table is that the majority of the fishermen (92%) as used traditional fishing. Remaining 8% of the respondents are used other artisanal.

4.15 Total Investment on Fishing

Table: 4.15

Total investment	No. of respondents	Percentage
Owned	44	88%
Borrowed	6	12%
Total	50	100%

Figure: 4.15



Source: Primary data

In the above figure shows that total investment on fishing inputs of the fishermen. 88% of the fishermen are owned fishing inputs and other 12% of the fishermen are borrowed fishing inputs.

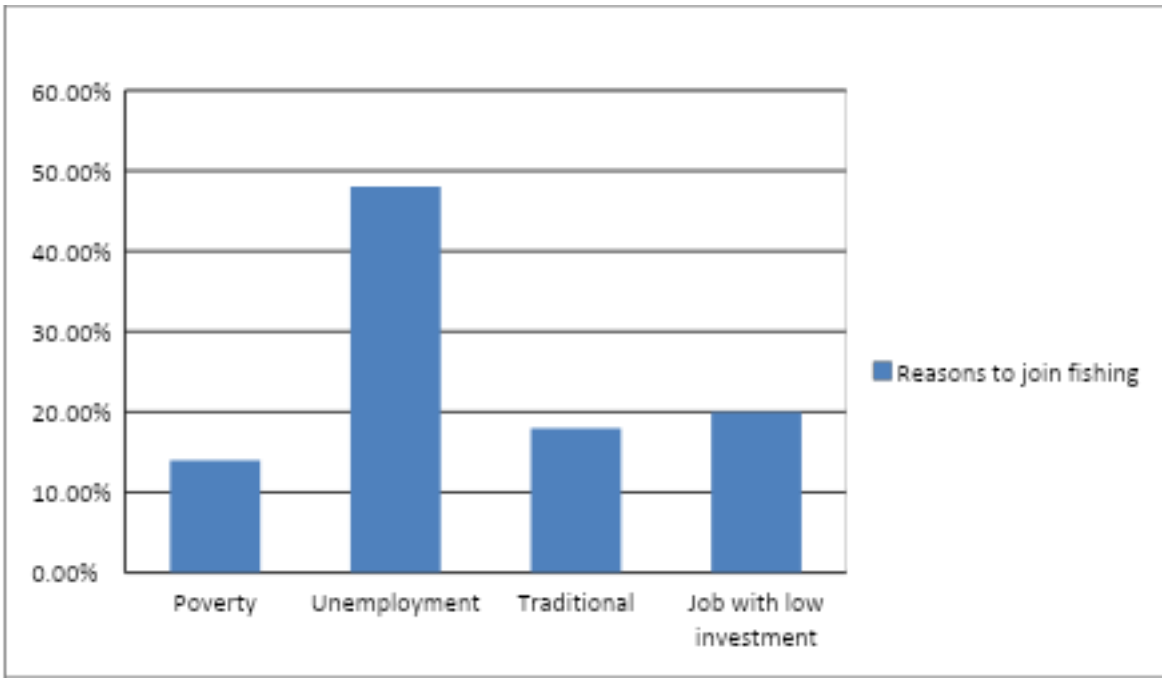
4.16 Reasons to join Fishing

The following table shows the different reasons to join the fishing.

Table: 4.16

Reasons to join fishing	No. of respondents	Percentage
Poverty	7	14%
Unemployment	24	48%
Traditional	9	18%
Job with low investment	10	20%
Total	50	100%

Figure: 4.16



Source: Primary data

From the above figure we can identify the respondents reasons to join fishing. Unemployment is the major reason for join fishing. Job with low investment is attracting to job less peoples to join fishing and 20% of the respondents are join the reason for job with low investment.18% of the people follow fishing traditionally and 14% of the people to join the reason is poverty. Poverty and unemployment is negatively affecting sample area.

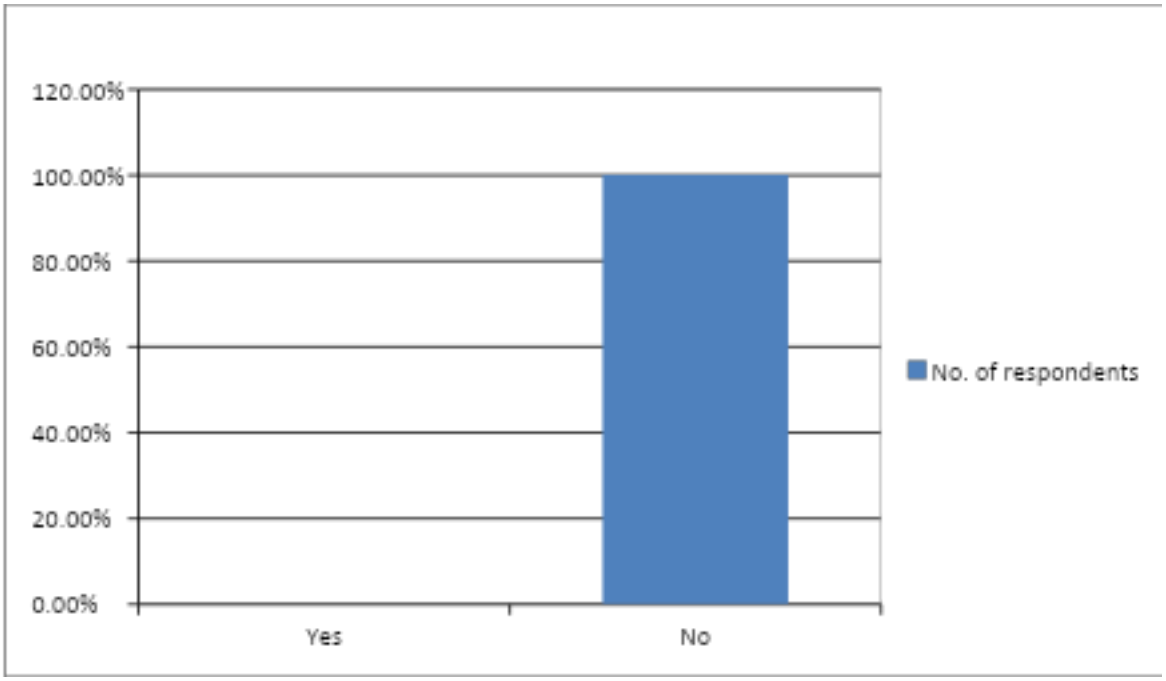
4.17 Children wish to join Fishing

Table: 4.17

Children wish to join fishing	No. of respondents	Percentage
YES	0	0
NO	50	100%

Total	50	100%

Figure: 4.17



Source: Primary data

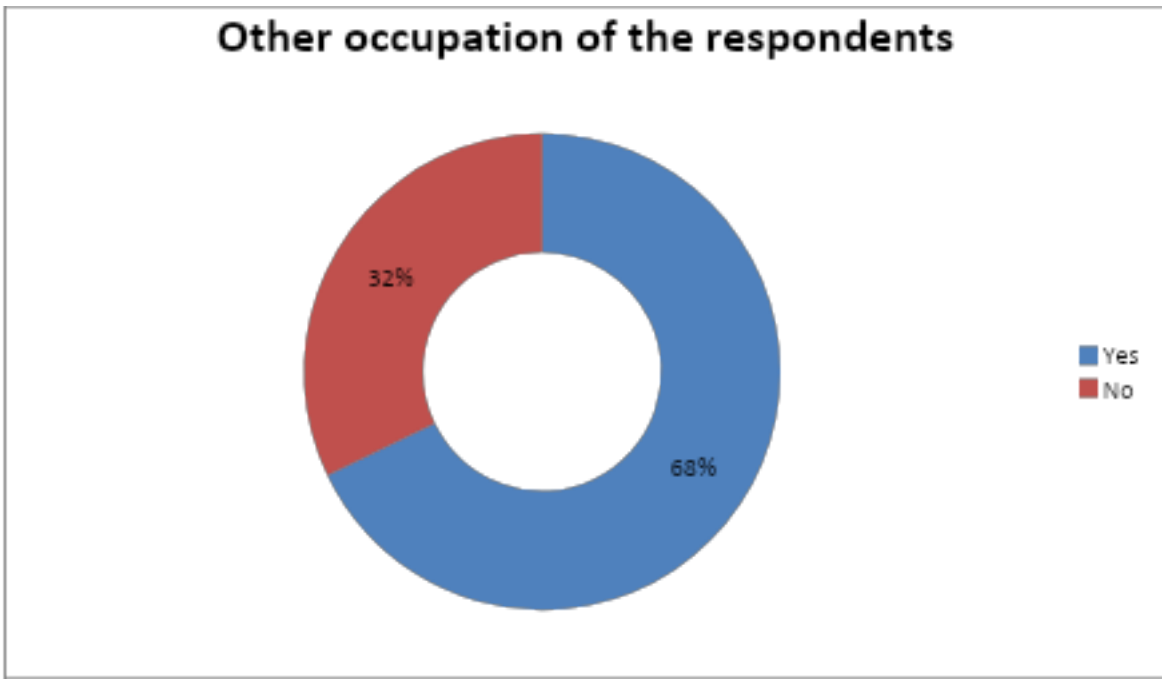
In the above figure as clearly no one can wish children join fishing.100% of the respondents are not supported by children join fishing. The reason for that they haven't any status in the society and fishing is risky job.

4.18 Other Occupation of the Respondents

Table: 4.18

Other occupation	No. of respondents	Percentage
Yes	32	76%
No	18	36%
Total	50	100%

Figure: 4.18



Source: Primary data

In the above figure shows that 32% of the respondent fishermen's doesn't have any other occupation and they are not practice any other occupation rather than fishing.68% of the respondents are practice other occupation than fishing. Fishing is depends upon availability of fishing. During fish depletion time, income is not certain. Because of this, people engage in other occupation like mainly daily wage works.

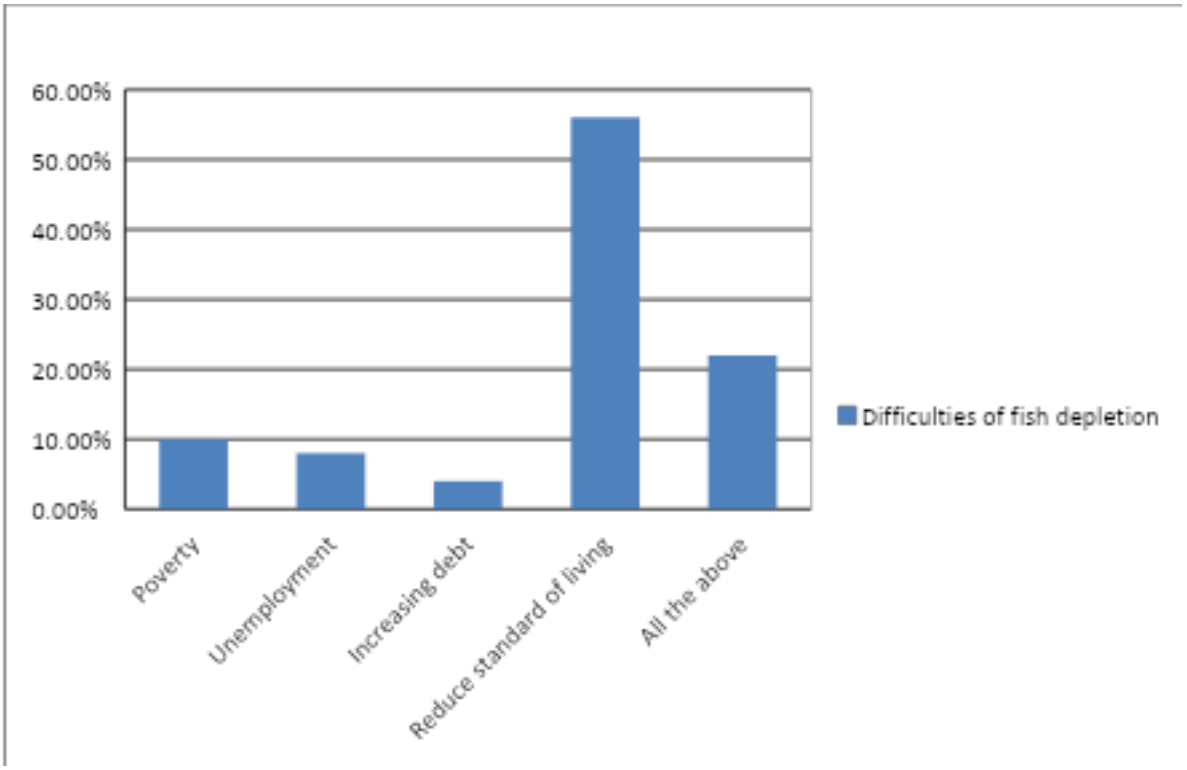
4.19 Difficulties of Fish depletion

Table: 4.19

Difficulties face fish depletion	No. of respondents	Percentage
Poverty	5	10%
Unemployment	4	8%
Increasing debt	2	4%
Reduce standard of living	28	56%

All the above	11	22%
total	50	100%

Figure: 4.19



Source: Primary data

In the above figure 56% of the respondents are face reduce standard of living during fish depletion. 22% of the respondents are face all the above difficulties like poverty, unemployment, increasing debt and reduce standard of living. In the time of fish depletion 10% of respondents face poverty, 8% of fishermen face unemployment and remaining 4% of the respondents are increasing their debts.

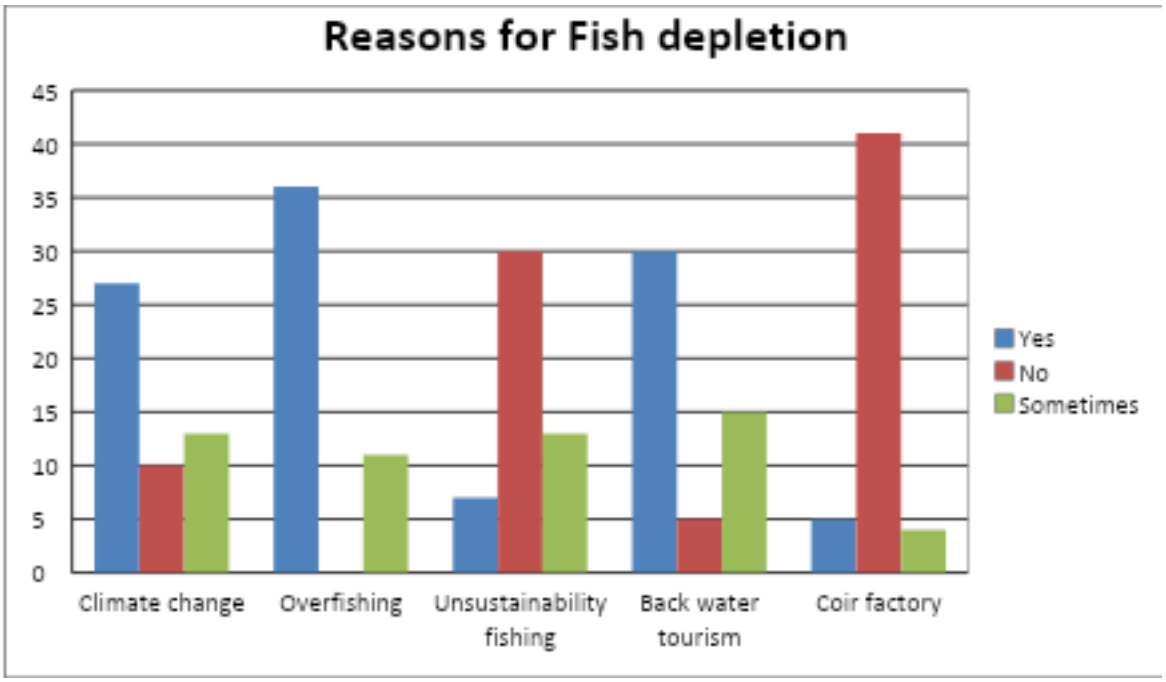
4.20 Reasons for Fish Depletion

Table: 4.20

Reasons for fish depletion	Yes	No	Sometimes	Total

Climate change	27	10	13	50
Overfishing	39	0	11	50
Unsustainability fishing	7	30	13	50
Back water tourism	30	5	15	50
Coir factory	5	41	4	50

Figure: 1.20



Source: Primary source

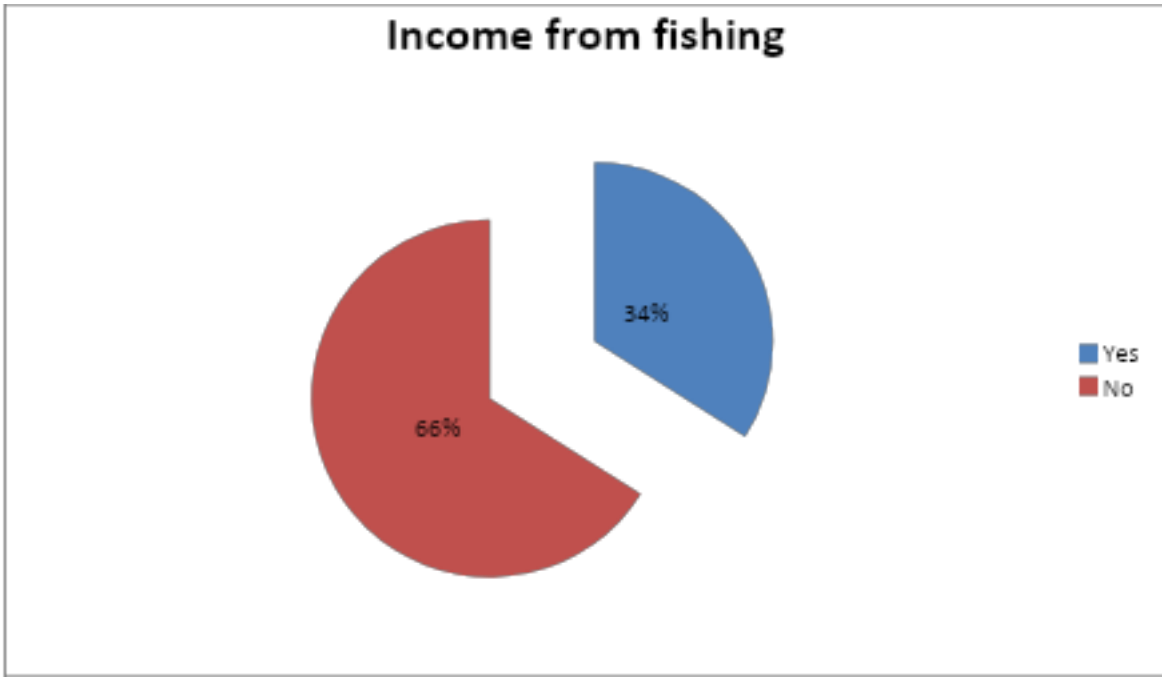
In the above figure shows that reasons for fish depletion. They have multiple reasons about fish depletion. Most of them suggested that overfishing is the reason for fish depletion. Another major reason for fish depletion is back water tourism and climate change.

4.21 Income from Fishing

Table: 4.21

Earning from fishing sufficient	No. of respondents	Percentage
Yes	17	34%
No	33	66%
Total	50	100%

Figure: 4.21



Source: primary data

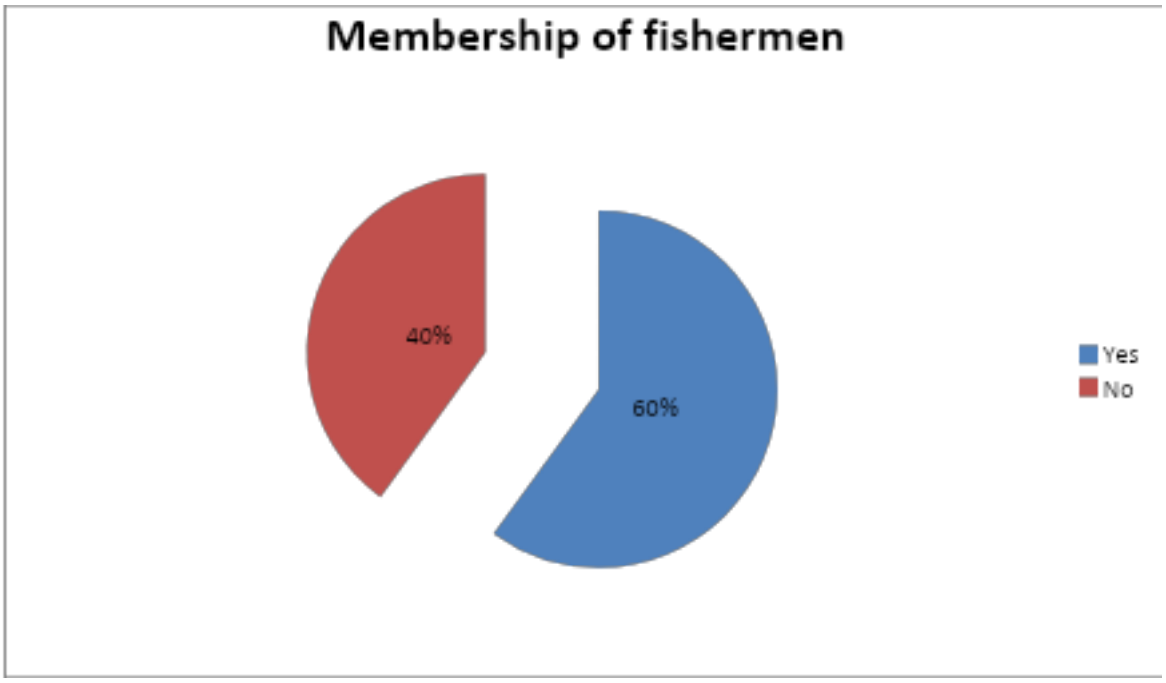
In this figure shows that 66% of the respondents earning from fishing is not sufficient and only 34% of the respondents income from fishing is sufficient. Most of the respondents depend on the other earnings like farming and other daily wage work.

4.22 Membership of fishermen in cooperative society

Table: 4.22

Member of any cooperative society	No. of respondents	Percentage
Yes	30	60%
No	20	40%
Total	50	100%

Figure: 4.22



Source: Primary data

In the above figure shows that most of the respondents (60%) have a member of fishing related cooperative society to get better price of fish and to avoid bureaucratic problems of the fishermen. Remaining 40% of the respondents are not a member of cooperative society.

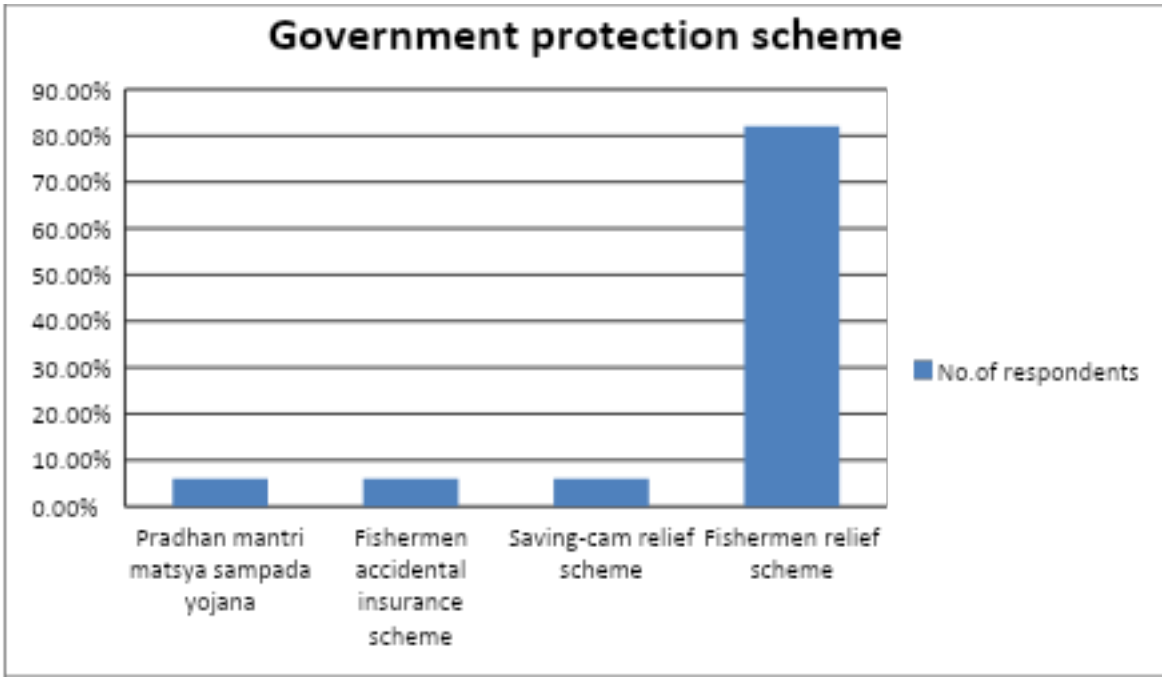
4.23 Respondents getting any government protection scheme in fishing

Table: 4.23

Government scheme	No. of respondents	Percentage

Pradhan mantri matsya sampada yojana	3	6%
Fishermen accidental insurance scheme	3	6%
Saving-cam relief scheme	3	6%
Fishermen relief scheme	41	82%
Total	50	100%

Figure: 4.23



Source: Primary data

In the above figure explains protection of the fishermen about services offered by government.82% of the fishermen to get fishermen relief scheme and each 3% of the fishermen get other protection scheme of the government.

CHAPTER 5

FINDINGS, SUGGESTIONS

AND CONCLUSION

INTRODUCTION

This survey was successfully conducted. The analysis of data is given in the previous chapter. Based on the data presentation and interpretation, important and relevant findings are identified. The findings of the study reveal information about fish stock depletion which threatens livelihoods on fisheries sector in Kerala. It also conveys and unveils the problems associated with fish stock depletion impact on fisheries sector in Kerala. The significant findings are given below.

5.1 FINDINGS

- From the analysis it is found that 96% of the respondents are male category and only 4% of the respondents are females.
- Most of the respondents are in the age group of 45-55 that is 34%. 30% of the respondents are in both the age group of 35-45 and 55-65. Only 6% of the respondents are in the age group of 25-35.
- About 72% of the respondents are Hindu community and remaining 28% of the respondents are Christian community.
- The 64% of the respondents are BPL category and remaining 36% of the respondents are APL category
- According to their caste 26 members are belonging to OBC and 11 respondents belong to general categories and 9 members belonging to SC. The ST categories are very less.
- The marital status of the respondents as 22% of the respondents are unmarried and 68% of them are married. 10% of the respondents are widower.

- About the educational qualification of the respondents, 46% of the respondents have completed SSLC. 38% of the respondents are higher educators and only 2% of them are illiterate.
- About 58% of the respondents family two earning members and 28% of the respondents family has three earning members and 10% of the respondents family only one earning member.
- The household members of the respondent family 78% of the household members are not work with the respondent. Only 8% of the two household members work with the respondent and 16% of the household one member is work with the respondent.
- About 30% of the fishermen receive an income below 50000 and 50% of the fishermen receive income between 50000-100000.18% of the respondents annual income is 100000-150000.only 2% of the fishermen receive the income of above 200000. The survey shows that thought all the members were not from the below poverty line family.
- The savings of the fishermen 64% is less than 50000. Only 10% of the respondents have between 50000-100000savings. 26% of the respondents are no savings.
- 28% of the respondents their loans and borrowings is 100000-150000and 26% of the respondents are 50000-100000. 26% of the respondents are no loans and borrowings.
- The 90% of the respondents have own land and 10% of the respondents are landless.
- The 56% of the respondents are land ownership is inherited and 28% of the respondents are purchased and 4% are given by the government. The remaining 12% are other ownership factors.
- The fishing assets of the respondents, most of the fishermen are used engine by fishing and 12 respondents are used other equipments.

- The 32% of the fishermen are 20-30 years experienced in fishing and 26% of the fishermen are well experienced.
- A notable fact is that the majority of the fishermen 92% as used traditional fishing and remaining 8% of the respondents are used other artisanal.
- In this study we can understand the majority of the respondents (48%) as unemployment is the major reason for join fishing and 20% as job with low investment is attracting job less people to join fishing. 18% of the respondents as follow fishing traditionally.
- It as clear no one can wish children to join fishing. The 100% of the respondents are not supported children to join fishing.
- The 32% of the respondent fishermen are not any other occupation and 68% of the respondents are practice other occupation rather than fishing. Fish depletion time, income is not certain. Because of this, people engage in other occupation like mainly daily wage works.
- During the fish depletion time 56% of the respondent family face decreasing standard of living and 22% all the difficulties like poverty, unemployment, increasing debt and reduce standard of living.
- They have multiple reasons about fish depletion. Most of them suggested that overfishing is the reasons for fish depletion. Other major reason for fish depletion is back water tourism and climate change.
- In this study 66% of the respondents earning from fishing are not sufficient and only 34% of the respondents' income from fishing is sufficient.
- The 60% of the respondents have a member of fishing related cooperative society to get better price of fish and to avoid bureaucratic problems of the fishermen. Remaining 40% of the respondents are not a member of cooperative society.

- The 82% of the fishermen to get fishermen relief scheme and each 3% of the fishermen get other protection scheme of the government.

5.2 SUGGESTIONS

- Government should introduce more welfare schemes for the development of fishermen.
- Regular orientation and training programmes should be conducted for fishermen in Panchayat level.
- To generate additional employment opportunities for the fisherman to help them alleviate poverty.
- Governments to take steps to increase fish production, quality of the fishes and improve nutritional content of the fishes.
- Increase awareness among fishermen in and around Vembanad on ethical fishing practices.
- Ensure that fisheries management and development interventions have a high likelihood of being sustainable in the long term.
- Provide more subsidies for small-scale fishing communities.
- To improve the health, social, financial and the economical conditions of the fishermen community.
- Fishermen must avoid alcoholic habits for better earnings.
- Fishermen have select modern technology for fishing.
- During the fish depletion time Government should arrange another job opportunities for fishermen community.
- Any non-wetland use of the lake should be strictly regulated.
- Organic pollution from untreated sewage from the nearby towns and house boats and resorts are a major reason of concern. The government should take utmost care not to open any untreated sewage into the lake.
- Strict measures should be adopted to prevent the deposition of plastic and other solid waste into the lake.

- Any development or modification works on critical fish habitats in Vembanad Lake should be done with proper impact assessment studies.

5.3 CONCLUSION

The purpose this study is to measure the existing level of socio-economic status of the fishermen in Kumarakom and to investigate the problems faced by fishermen due to fish stock depletion. In the time of fish stock depletion, fishermen in Kumrakom point out they faced by problem is reduce standard of living, poverty and unemployment. So these time to take steps to increase fish production, quality of the fishes and improve nutritional content of the fishes. Income revenue, create more job opportunity, reduce risks in fishing, and develop fisheries infrastructure, improve the quality of the rural life of these measure can solve the livelihood of fishermen and make their lives easier.

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APPENDIX

QUESTIONNAIRE

A STUDY ON FISH STOCK DEPLETION WHICH THREATEN LIVELIHOODS ON FISHERIES SECTOR IN KERALA WITH SPECIAL REFERENCE TO VEMBANABU LAKE, KUMARAKOM

1. Name :
2. Age : 25-35 ☐ 35-45 ☐ 45-55 ☐ 55-65 ☐
3. Gender : Male ☐ Female ☐ Transgender ☐
4. Religion : Hindu ☐ Christian ☐ Muslim ☐
5. Caste : SC ☐ ST ☐ OBC ☐ Other ☐
6. Household income status: APL ☐ BPL ☐
7. Marital status : Married ☐ Unmarried ☐ Widower ☐
8. Educational qualification: Illiterate ☐ Below SSLC ☐ SSLC ☐ Graduate ☐ Others ☐
9. No. of household earners:
10. Number of household members working with you?

    One ☐ Two ☐ Not at all ☐

11. Dependents:
12. Annual income:

    a. Below Rs.50,000 ☐ b. Rs. 50,000-100,000 ☐

    c. Rs. 100,000-150,000 ☐ d. Above Rs.200, 000 ☐

13. Savings:

    a. Below Rs.50,000 ☐ b. Rs. 50,000-100,000 ☐

    c. Rs.100,000-150,000 ☐ d. Above Rs. 200,000 ☐

14. Loans and borrowings

a. Below Rs. 50,000 ☐ b. Rs.50,000-100,000 ☐

c. Rs. 100,000-150,000 ☐ d. above Rs. 200,000☐

15. Land ownership: Yes ☐ No ☐

16. If, Yes mode of inheritance

a. Inherited ☐ b. Purchased ☐ c. Granted by Government ☐

d. Others ☐

17. Fishing Assets

a. Craft : Yes ☐ No ☐

b. Engine : Yes ☐ No ☐

c. Gear : Yes ☐ No ☐

d. Hook and line: Yes ☐ No ☐

e. Other equipment: Yes ☐ No ☐

18. How long have you been to fishing:

19. Which fishing category you belong to?

a. Traditional ☐ b. Other artisanal ☐

20. Total investment on fishing inputs?

a. Owned ☐ Borrowed ☐

21. Specify reason to join fishing?

a. Poverty ☐ b. Unemployment ☐ c. Traditional ☐

d. Job with low investment ☐

22. Do you children wish to join fishing: Yes ☐ No ☐

23. Do you practice any other occupation other than fishing?

Yes ☐ No ☐

24. What are the difficulties that you face during the fish depletion?

a. Poverty ☐

b. Unemployment ☐

c. Increasing debt ☐

d. Reduce standard of living ☐

e. All the above ☐

25. Reasons for fish depletion:

Reasons for fish depletion	Yes	No	Sometimes
Climate change			
Overfishing			
Unaustainability of fishing			
Back water tourism			
Coir factory			

26. Is the earning from fishing sufficient?

Yes ☐ No ☐

27. Do you think that continuing in fishing is feasible in the future?

Yes ☐ No ☐

28. Are you a member of any of the cooperative related to fishermen?

Yes ☐ No ☐

29. Why did you join cooperatives?

a. To get government aids ☐

b. To get better price ☐

c. To avoid bureaucratic problems ☐

d. Compulsory ☐

e. All the above ☐

30. Has your income levels increased being a fishermen?

Yes ☐ No ☐

31. Are you getting any government protection scheme in fishing?

a. Pradhan mantri matsya sampada yojana

b. Fishermen accidental insurance scheme
c. Saving-cam relief scheme
d. Fishermen relief scheme